Corrections and Clarifications
FFGFT / T0 Time-Mass Duality

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Contents

Preliminary Note

Status: C1–C6, P1–R43 (P16/P17 partial, P20/P30/P32 open, P29 partial, P31 partially derived, P33 declared, P34 clarified; P35/P36 added; P37 revised; C4 (factor 3) added, C5/P38 after external audit; P39 H_0/R_H status clarified; C6 redshift achromatic; P40 absolute values carry correction via bridge constants; R41–R43 lepton sector/calibration) — Register R1–R74, C1–C7. *Status 5 August 2026:* R74 (lithium problem in 025/063 as a mechanism, not a solution; robustness classification from Doc. 313). *Status 3 August 2026:* R67 (accuracy of the factor 100, ± 2), R68–R73 (clarifications arising from work on Docs. 312/313: Λ^* as carrier of the Λ function and full adoption of the field equations [279/308], a_0 via the compact time direction [308], T_0 scale gap and notation clash [061/A085], side-taking in the Hubble tension [309], E_0 value [A010/A080/A265], H_0 notations and exponent $41/4$ [026/279]).

Legend (layer labels). Every substantive claim carries one of the following labels in square brackets:

[K] *Core derivation* — derived mathematically or numerically from the foundational posits; internally consistent and reproducible.

[S] *Structural claim* — algebraically or geometrically motivated, but not yet fully proved.

[POSIT] *Posit* — declared assumption or definition not further derived.

[B] *Motivated* — geometrically motivated but no formal proof; between [S] and [POSIT].

P-numbers denote open programme points (gaps still to be closed or calculations in progress); **C-numbers** are completed corrections or anchored findings; **R-numbers** are revisions of previously booked claims. The numbering runs continuously across the letters; the letter marks the type, the number is global.

The corrections (C1–C3) and refinements (P1–P13) listed in this document have been incorporated into the respective source documents. The last open item — C2 (tau prefactor r_τ) — was fully completed on 27 May 2026: besides the items reported on 26 May in Doc. 016, Doc. 046 and Doc. 116, this also affected remaining *derived* values in Doc. 046 DE (energy $18{,}8 \to 18{,}0$ meV) as well as the entire **English** parallel version of Docs. 016, 046 and 116, which had not yet been updated on 26 May. For details and the target state see the section "Status of incorporation" under C2.

This document is retained as a historical archive. For r_τ the value $25/9$ recorded here is authoritative.

Addendum (2 June 2026): Added and incorporated is **P14a** (formulation of the redshift in Doc. 041): implemented in Doc. 041 (DE+EN).

Addendum (6 June 2026): Carried over from the CMB work (Doc. 267 Cosmological Degeneracy, Doc. 268 CMB peaks from T^4): **P18** (ΛCDM/FFGFT as two readings, theoretical not empirical), **P20** (absolute cosmological scale: *form* fixed as the ξ ratio R_H/ℓ_P, the *factor* $41/4$ a declared external calibration from ΛCDM, not derived, open), **P29** (peak selection rule, only *partially* answered – the factor-3 filter does *not* exclude $|n|^2 = 30$), **P30** (rigorous C_ℓ projection; the naive spherical Bessel sum is *insufficient*, open) and **P31** (geometric anharmonicity from the codimension-1 projection – leading behaviour derived forward, small residual open). New **P32** (statements presenting H_0 as FFGFT-derived are to be cleaned up corpus-wide – noted for now). The previously separate items *P19* (CMB path lengthening) and *P28* (exponent for z_*) are merged into **P14a** resp. **P20** and do not appear as their own rows.

Addendum (8 June 2026): New **P33** – the magnitude-carrying factor 5^4 in $\xi = 1/(3 \cdot 2^2 \cdot 5^4)$ is *not* derived forward but grounded intuitively (harmonic / Euler Tonnetz) resp. empirically (Higgs sector); the *form* ($4/3$, 3, 2^2) is forward, the *magnitude* (5^4) remains at the intuitive derivation and is openly *declared* as such. Structural parallel to P20 (form fixed, factor open).

Addendum (8 June 2026, II): New **P34** (dimensional structure). Doc. 181 presents $D = 4$ as “following necessarily”; clarified: **3+1 is empirical input** (no theory derives the dimension count). Forward are only (i) periodicity, (ii) torus-not-sphere ($\pi_1 \neq 0$), (iii) minimal sufficiency (4 instead of 5 via the duality $\tilde{T}m = 1$), (iv) the lossless Hilbert representation (Type III). At the same time recorded: the SI conversions are *indispensable* for the exponent structure (*power of* c, power of time) – c is the space-time bridge (T^3 ↔time), $\hbar$ the mass-time bridge (S^1_m); $c = \hbar = 1$ collapses L, T, M and destroys the powers. The four-circle structure *is* this bookkeeping. Signature as P20/P33 (structure forward, number input).

Addendum (10 June 2026): New **P37** (use of φ in various documents – clarification of the levels). Occasion: Doc. 275 shows that $1/\varphi$ is the fixed point of the ξ damping recursion after $k^* = \log(\varphi)/\xi \approx 3609$ scale levels. In the process it turned out that φ appears in the corpus in *three different roles* that must not be conflated. P37 documents this separation and augments the ξ^N/φ^N triviality warning (P35) by the role level. Note: P35 (ξ^N triviality) and P36 (R_H interpretation) had been declared via the changelog (Updates 16/17); their table entries were added on 10 June 2026.

Addendum (11 June 2026): New **C5** (three implementation errors in the validation script v3 for Doc. 275: point duplicates, seed bug, detector bias – all fixed in v4, affected results retracted) and **P38** (k^* result downgraded from “fixed point without a free parameter” to “pass-through point, P35 applies”; P37 level (2) revised accordingly). Both entries go back to the independent audit by P. Stoychev (IPI list, 11 June 2026) – the first external audit entry of the register.

Addendum (14 June 2026): New **P39** (status of H_0/R_H clarified). Occasion: Doc. 277. It is clarified that the relation Planck length↔ R_H (exponent $41/4$) follows from *no* framework out of first principles; ΛCDM fixes it via the measured H_0 (itself circular, Doc. 267), FFGFT adopts the established value for comparability. Hence the cosmic scale is a *shared empirical reference point*, not an FFGFT-specific gap and not an asymmetric disadvantage relative to ΛCDM (P20/P32/P36 remain valid but are no longer read as a weakness). The only remaining open point is the forward derivation of $41/4$ from ξ – symmetric, since no framework has it; the only model asymmetry is the dark sector (carried by ΛCDM alone). At the same time a numbering conflict was resolved: the v3 script audit (initially recorded as C4 in the addendum of 11 June) is now **C5**; **C4** denotes the older factor-3 correction (Doc. 267/268, 3D observer instead of fractal dispersion), added as a table row.

Addendum (15 June 2026): New **C6** (redshift is *achromatic*, not wavelength-dependent). Occasion: Doc. 280 and the finite-element simulation of the fractal path-lengthening. It corrects the earlier chromatic presentation ($z(\lambda) \propto \lambda$, “UV stronger than radio”) in Doc. 004, 025, 030, 039, 041, 053, 061, 063, 068, 081. The mature mechanism (geometric path-lengthening) stretches all wavelengths by the same factor; the observed cosmological redshift is achromatic, which FFGFT gets right. The earlier “wavelength-dependent discriminator” drops out – it moves to the “fits” column (data degeneracy, Doc. 267).

Addendum (15 June 2026): New **P40** (absolute values carry the fractal/SI correction via the bridge constants; ratios are exact). In FFGFT only the *dimensionless ratios* are exact. Absolute values carry the fractal/SI correction because it is absorbed into the dimensionful *bridge constants*: v (masses $m = r\xi^p v$), $E_0 = 7{,}398$ MeV ($\alpha = \xi E_0^2$) and the factors $3{,}521 \times 10^{-2}/2{,}843 \times 10^{-5}$ (G; canonical script `calc_De.py`, which carries *no* explicit K). In ratios v/E_0/factors cancel ($m_\mu/m_e = 206{,}768$, correction-free). Factored out, this correction is $K_{\text{frak}} = 1 - 100\,\xi \approx 0{,}987$ (Doc. 011/012; in Doc. 133 derived from D_f-consistency and an RG run over 17 orders of magnitude – *not an adjusted correction*). **Do NOT confuse two numbers:** the *100* in K_{frak} is the RG amplification of the $\sim 1{,}3\,\%$ correction (Doc. 133); the *3609* (Doc. 275) is the depth of the ξ-scale recursion $r(k) = \frac{D_f}{3}(1-\xi)^k$ at which the sequence *passes* $1/\varphi$. There $1/\varphi$ is a *passage point, not a fixed point* (the only fixed point is 0); since the sequence passes *every* value, 3609 is unanchored unless $1/\varphi$ is independently motivated (status open, P37). Moreover further mutual couplings are not excluded (⇒ even ratios are leading order; *backdoor*): a deviation from the simple rational can always be attributed to an unmodelled coupling – ratio tests confirm (Koide 10^{-5} at $2/3$) but barely falsify. Clean falsification only via a *forward*-fixed new prediction ($41/4 \to H_0 \approx 66{,}2$ against the H_0 tension).

Addendum (2 July 2026): New **P41** (redshift drift Sandage–Loeb checked and not booked: amplitude test not degeneracy-proof at the shared scale $c/R_H = H_0$, only a conditional shape test) and **P42** (lepton sector: one declared reference point m_μ/m_e; $2/9$ as the rational address of θ_{ref}, sharpened prediction $m_\tau = 1776.9690$ MeV; occasion: precision analysis Doc. 292). At the same time the doubly assigned heading "Refinement 14" was resolved into **P14a** (redshift Doc. 041, incl. table row) and **P14b** (CMB anharmonicity, covering P29–P31).

This document lists corrections and refinements to already published documents of the FFGFT series. Older documents are *not* changed. Where a contradiction exists between an older and a newer document, the version recorded here is authoritative.

Each entry contains: the affected document and location, the erroneous expression, the correct expression, and a short justification.

.1 Correction 1: Prefactor in the Higgs derivation of ξ

Affected document

Doc. 049 (T0 Lagrangian and Higgs integration), HTML interim version and older working versions.

Erroneous expression

$$\xi = \frac{\lambda_h^2 v^2}{64\pi^4 m_h^2} \approx 1{,}0 \times 10^{-5} \qquad \text{(old – wrong)}$$

Correct expression

$$\boxed{\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1{,}33 \times 10^{-4}} \qquad (190.1)$$

with the experimental inputs:

$$\begin{aligned} m_h &= 125{,}25\,\text{GeV} && \text{(Higgs mass)},\\ v &= 246{,}22\,\text{GeV} && \text{(vacuum expectation value)},\\ \lambda_h &= \frac{m_h^2}{2v^2} \approx 0{,}129 && \text{(Higgs self-coupling)}. \end{aligned}$$

Justification

The prefactor $64\pi^4$ stems from a typo in the early HTML visualization. It yields $\xi \approx 10^{-5}$, which deviates by an order of magnitude from the geometrically derived value $\xi = 4/3 \cdot 10^{-4}$ and does not reproduce the natural constants. The correct prefactor $16\pi^3$ gives $\xi \approx 1{,}33\times10^{-4}$, consistent with the independently determined value from the proton-electron mass ratio (Doc. 009).

.2 Correction 2: Tau Yukawa prefactor r_τ

Affected documents

The prefactor r_τ appears in several documents. Complete list:

- Doc. 116 (Koide formula as a consequence of T0 theory), lepton-parameter table.
- Doc. 016 (Complete calculations), mass table and r-parameter list.
- Doc. 046 (Particle masses), tau-neutrino row (double-ξ construction containing r_τ).

Already correctly updated at the time of the first correction were Docs. 006, 028, 158, 210 as well as the theorem and proof part of Doc. 116.

Erroneous expression

$$r_\tau = \frac{8}{3} \approx 2{,}667 \quad \Rightarrow \quad m_\tau \approx 1712\,\text{MeV} \quad (-3{,}6\,\%\ \text{dev.}) \qquad \text{(old – wrong)}$$

Correct expression

$$\boxed{r_\tau = \frac{25}{9} \approx 2{,}778} \quad \Rightarrow \quad m_\tau \approx 1783\,\text{MeV} \quad (+0{,}4\,\%\ \text{dev.}) \tag{190.2}$$

The lepton masses follow generally from:

$$m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v \tag{190.3}$$

with the exponents $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$ (step size $\Delta p = 1/3$) and the vacuum expectation value v.

Justification

$r_\tau = 8/3$ was a remnant from an earlier iteration step. Doc. 158 (Koide-to-g-2 bridge) already uses $r_\tau = 25/9$, which matches the tau mass numerically. Moreover $25/9 = (5/3)^2$ has a geometric interpretation as the square of the pure sixth in the natural harmonic series, consistent with the musical resonance analogy (Doc. 060).

Complete corrected prefactor table

Lepton	Exponent p	Prefactor r	m [MeV]
Electron	$3/2$	$4/3$	0,511
Muon	1	$16/5$	105,7
Tau	$2/3$	$25/9$	1783

Status of incorporation (Addendum 26 May 2026)

A renewed corpus-wide search found **remaining $8/3$ residues**. The correction had thus *not been carried through cleanly*; the original entry named only Doc. 116 as affected and overlooked Docs. 016 and 046.

Doc. 116, lepton-parameter tab. Finding: $r_\tau = 8/3$; Target: $25/9$ contradicts its own theorem

Doc. 016, mass tab. and r-list Finding: $r_\tau = 8/3$; Target: $25/9$.

$m_\tau = 1712{,}1$, $-3{,}64\,\%$ Target: ≈ 1783, $+0{,}3\,\%$.

Doc. 046, tau-neutrino (3 places) Finding: $r = 8/3$; Target: $25/9$ (param. row, double-ξ calc., second ν table; cf. Doc. 006)

Carried out on 26 May 2026: All of the above places were brought to $25/9$; in Doc. 016 the derived tau mass ($1712{,}1 \to 1783{,}4$ MeV) and the deviation ($3{,}64 \to 0{,}37\,\%$) were recomputed, in Doc. 046 the ξ_{ν_τ} calculation ($128/27 \to 400/81$, $E = 18{,}8 \to 18{,}0$ meV). C2 is thereby fully incorporated.

Addendum 27 May 2026 (actual extent): An independent full check on 27 May found that the incorporation on 26 May was *not* complete. Two points were still open:

- **Doc. 046 DE** still carried, in the *derived* tables (compatibility, method-equivalence and falsification tables), the old energy value $18{,}8$ meV as well as the decimal form $r_\tau = 2{,}768$. The main calculation ($\xi_{\nu_\tau} = 400/81$) was already correct; only the downstream values had not been updated. Corrected to $18{,}0$ meV resp. $2{,}778$.
- The **English parallel version** of Docs. 016, 046 and 116 had *not* been updated at all on 26 May (DE and EN must be consistent). Corrected: 016 EN (mass table $8/3 \to 25/9$, $1712{,}1 \to 1783{,}4$, $3{,}64 \to 0{,}37\,\%$; r-list), 046 EN (both ν_τ rows, ξ_{ν_τ} calc. $128/27 \to 400/81$, three E values, method equivalence), 116 EN (tau row $8/3 \to 25/9$).

Corpus-wide residual search afterwards: **0 remaining** $8/3$ **places** in 016/046/116 (DE+EN). All four files recompiled (Kindle 6×9). C2 has been *fully* incorporated since 27 May 2026 — DE and EN.

.3 Correction 3: Base formula for g-2 of the electron

Affected document

T0-Explorer (HTML visualization), first version; older interim versions of Doc. 018.

Erroneous expression

$$a_e = \frac{4\pi}{f \cdot k_{\text{geom}}} \qquad \text{(old – wrong)}$$

Correct expression

$$\boxed{a_e = \frac{S_3}{f \cdot k_{\text{geom}}} = \frac{2\pi^2}{f \cdot k_{\text{geom}}}} \tag{190.4}$$

with $f = 1/\xi \approx 7500$, $k_{\text{geom}} = (2/\sqrt{\varphi}) \cdot \sqrt{2} \approx 2{,}224$ and $S_3 = 2\pi^2 \approx 19{,}74$ (surface of the 4D unit sphere, i.e. of the 3D boundary of the torus).

Justification

The earlier prefactor 4π corresponds to the surface of the 2D sphere in 3D space, not to the geometrically correct 3D surface of the 4D torus. $S_3 = 2\pi^2$ is the correct value from the torus geometry. The fractal corrections for muon and tau continue to use 4π:

$$\Delta a_\mu = \frac{4\pi}{f^{5/3}}, \tag{190.5}$$

$$\Delta a_\tau = \frac{4\pi}{f^{4/3}}. \tag{190.6}$$

.4 Refinement 1: Accuracy of the Koide formula

Affected documents

Doc. 116 (abstract, l. 14): $\Delta Q < 0{,}00003\,\%$
Doc. 158 (abstract): "experimentally confirmed to $0{,}001\,\%$"

Correct placement

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{190.7}$$

is experimentally confirmed to better than $0{,}001\,\%$ (PDG values 2022). The figure $< 0{,}00003\,\%$ in Doc. 116 refers to the internal consistency of the T0 formulas, not to the experimental measurement uncertainty. Both statements are correct in content but are different quantities. In external communication $0{,}001\,\%$ is the appropriate figure.

Note

The T0 individual masses agree with experiment to $\sim 1\,\%$; the high precision of $Q = 2/3$ arises from the specific algebraic structure of the Koide combination, not from individual mass accuracy.

.5 Refinement 2: $\hbar$ from the Higgs field – derivation chain

Context

The Planck constant $\hbar$ is derived differently in various documents. Here the complete, consistent derivation chain is recorded.

Authoritative derivation chain

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \longrightarrow L_0 = \xi \cdot \ell_P \longrightarrow \hbar \sim \sqrt{\xi} \cdot \ell_P \cdot c \tag{190.8}$$

The time-mass binding condition provides the physical justification:

$$T(x,t) \cdot m(x,t) = 1 \quad \Longrightarrow \quad T = \frac{\hbar}{m c^2} \quad \Longrightarrow \quad E = \hbar\omega. \tag{190.9}$$

$h = 2\pi\hbar$ is thereby *not* a free constant but a geometric consequence of ξ and ℓ_P. The numerical SI value $h \approx 6{,}626 \times 10^{-34}$ J·s is the calibration result of this relation onto the SI base units.

.6 Refinement 3: Two ξ parameters – two different spaces

Context

In Docs. 173–176 (quantum dots, spin qubits, qubit state spaces, Shor algorithm) two numerically different ξ values appear. Earlier documents did not keep them consistently separate, which can cause confusion.

Authoritative distinction

ξ_0 **(geometric)** Value: $\frac{4}{30000} \approx 1{,}33 \times 10^{-4}$; Domain: 3D geometric space; Meaning: length hierarchy of the torus; holds in physical position space.

ξ_{Higgs} **(algebraic)** Value: $\approx 1{,}04 \times 10^{-5}$; Domain: number / quantum state space; Meaning: algebraic field corrections; Higgs sector; decoherence rates.

Justification

$\xi_0 = 4/30000$ follows from the torus geometry of physical space (geometric derivation, independent of SI measurements). ξ_{Higgs} follows from the algebraic Higgs field structure and appears in coupling formulas between quantum states (e.g. Bell correlations, T_2 hierarchy).

The two ξ values are not a contradiction, but describe complementary aspects: geometry of space (ξ_0) vs. algebra of the state space (ξ_{Higgs}). In external communication it must always be made clear which parameter is meant.

.7 Refinement 4: L_0 and the Schwarzschild interpretation

Affected documents

Older versions of Docs. 180–182 (Lagrangian derivation, torus geometry, maximal universe scale).

Erroneous approach

Earlier versions interpreted $L_0 = \xi_0 \cdot \ell_P$ as the Schwarzschild radius of a fundamental minimal entity. This interpretation was explicitly discarded in Doc. 182 (2026 revision).

Correct placement

L_0 is exclusively the **geometric fundamental length** of the FFGFT torus:

$$L_0 = \xi_0 \cdot \ell_P \approx 2{,}15 \times 10^{-39}\ \text{m}. \tag{190.10}$$

It defines the lowest scale level of the ξ hierarchy and has *no* Schwarzschild interpretation. The cosmological Hubble radius R_H is the system-specific upper bound of the cosmological sector – not a universal maximal size. Every scale has its own system-specific upper bound.

Remark: Schwarzschild as a consistency check

The Schwarzschild connection remains valid as an **internal consistency check**: an object with mass $M_0 = \xi_0 \cdot m_P/2$ would have a Schwarzschild radius of exactly L_0 – without a free parameter. If M_0 matches a scale level of the ξ hierarchy, that is a nontrivial self-consistency test. The connection is *a probe, not a derivation*.

.8 Refinement 5: Claim and scope of the mass formulas

Context

Several FFGFT documents describe mass derivations for leptons and other particles. The claim of these derivations must be communicated clearly.

Authoritative formulation

The FFGFT mass formulas (e.g. $m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v$) yield values that agree with experiment to $\sim 1\,\%$. This is **not a precision calculation** but a structural proof:

- The mass formulas *follow geometrically* from ξ and v – they are not fitted.
- The remaining $\sim 1\,\%$ deviations reflect measurement uncertainties and approximations, not errors of the structure.
- Numerically better calculations than the Standard Model are *not* claimed.

In external texts the following is to be avoided: "FFGFT computes the masses exactly." Correct is: "FFGFT *derives* the mass structure from a single parameter ξ and reproduces the orders of magnitude to $\sim 1\,\%$."

.9 Refinement 6: Koide formula only with original values

Authoritative rule

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{190.11}$$

may only be evaluated with experimentally measured masses (PDG values).

Not admissible: inserting the symbolic FFGFT terms $r_\ell \cdot \xi^{p_\ell} \cdot v$ directly into the formula. The $\sqrt{m_i}$ operation generates cross terms with mixed ξ powers that are foreign to the torus structure space. Any such insertion yields $Q \approx 0{,}6677 \neq 2/3$ and is not a contradiction to the theory but a category error.

Admissible: evaluate the FFGFT terms numerically (e.g. $m_\tau = 1783$ MeV) and insert this result into the Koide formula like a measured value.

Mathematical background: The r_ℓ terms are geometric invariants of their torus modes. The set of admissible pairs (r_i, p_i) is not closed under multiplication: $r_e \cdot r_\mu = 64/15$ belongs to no torus mode. This is the non-closure theorem (Doc. 193). The general analysis of the composition limits is found in Doc. 192.

.10 Refinement 7: Terminology "renormalization group" and scale structure

Affected documents

- Doc. 005 (T0-tm extension): "RG flow", "renormalization-group factor", "renormalization-group invariance", subsection "renormalization-group treatment".
- Doc. 008 (ξ and e): subsection "modified renormalization-group equations".
- Doc. 019 (T0 Lagrangian, older version): subsection "renormalization-group equations".
- Doc. 086 (document overview): "renormalization group as a flow in parameter space".
- Doc. 189 (torus derivation): "cumulative renormalization-group running".
- Doc. 202 (FFGFT field theory, overall), §18: section title "The renormalization group".

Previous formulation

In the documents named, the ξ_0-modified scale running of the coupling

$$\frac{d\alpha}{d\ln\mu} = \beta_0\,\alpha^2\Big(1 + \xi_0\,\ln\frac{\mu}{E_0}\Big) \qquad \text{(old – imprecise)}$$

and the associated scale dependence are called the *renormalization group*, at times in immediate proximity to statements that characterize ξ_0 explicitly as a topologically fixed geometric constant.

Refined formulation

In FFGFT the scale dependence of the coupling is *not* the result of a renormalization procedure but a direct consequence of the recursion $\mathscr{R}$ (Doc. 203) on the fractal torus geometry. The correct designation is therefore:

$$\boxed{\begin{array}{c}\text{scale structure from the recursion}\\ \equiv\ \text{recursive } \alpha \text{ running via } \mathscr{R}\end{array}} \qquad (190.12)$$

Concretely:

- The term $\xi_0\,\ln(\mu/E_0)$ arises from the iterated application of the recursion $\mathscr{R}$ to the mode structure, not from a counter-term subtraction.
- ξ_0 is *not* a fixed point of a β function but a topologically fixed constant ($\xi_0 = 4/30000$, Doc. 180).
- Stability, scale flow and UV finiteness are three aspects of the *same* recursive structure – not separate mechanisms.

Language convention

renormalization group (implies a QFT procedure) Use: scale structure from the recursion.
RG flow / RG running Use: recursive α running; $\mathscr{R}$-induced scale correction.
renormalization-group equations Use: scale equations from $\mathscr{R}$.
renormalization-group invariance Use: recursive scale invariance.
renormalization-group fixed point Use: fixed point of the recursion: $\mathscr{R}(\Phi_*) = \Phi_*$.

Justification

The designation "renormalization group" imports a tool of standard quantum field theory that rests on a completely different principle: there, UV divergences are absorbed by a procedure, and the parameters *run* under a formal scale transformation. In FFGFT neither the divergence problem (cf. Doc. 159, "without artificial renormalization") nor a free scale parameter to be absorbed exists. The scale dependence is a *structural property* of the recursion, not a subsequent adjustment step.

The FFGFT-native view is already correctly formulated in Doc. 159 ("renormalization as a symptom, not a solution") and Doc. 189 ("geometrically determined, not by renormalization"); this refinement serves the uniform terminology across the older Lagrangian and ξ-derivation series (Docs. 005, 008, 019) as well as across Doc. 202.

The underlying mechanism is worked out in Doc. 203 (recursion operator $\mathscr{R}$) and Doc. 204 (fractal boundary condition).

.11 Refinement 8: Terminology "fractal renormalization"

Affected documents

Docs. 005, 008, 012, 014 (with 19 occurrences the most frequent source), 041, 060, 133, 141, 159, 181.

Previous formulation

"Fractal renormalization" is used as an FFGFT-internal term for the geometric correction factors in the conversion between natural and SI units as well as for scale-dependent corrections on the fractal torus.

Refined formulation

Since the word "renormalization", even in compound form, carries the QFT connotation of a subtraction or adjustment procedure, the term is replaced by context-specific alternatives:

unit conversion natural ↔ SI (esp. Doc. 014) Authoritative designation: geometric scale adjustment.

scale-dependent corrections on the fractal torus (Docs. 133, 159, 181) Authoritative designation: fractal correction (matches the title of Doc. 133).

general scale transformation by recursion Authoritative designation: $\mathscr{R}$-induced scale correction.

Justification

The internal term "fractal renormalization" was meant to set the procedure clearly apart from QFT renormalization, but it adopts the central word and with it inevitably the connotation of a *correction procedure*. In FFGFT it is not a procedure but the direct numerical manifestation of the fractal geometry. The proposed alternatives avoid this misunderstanding without new terminology – both phrasings are already established in the corpus (Doc. 133 "fractal correction"; "geometric scale adjustment" in the same semantic field as Doc. 014).

.12 Refinement 9: Column "symmetry" and the term "color charge" in the fermion tables

Affected documents

- Doc. 006 (T0 particle masses), "universal table of fermions", column "symmetry".
- Doc. 046 (Particle Masses, English parallel version), same table, column "color".
- Doc. 042 (ξ parameter and particles), table "spatial field patterns", entry "quarks: multi-node bound clusters, confined, color charge".
- Doc. 005 (T0-tm extension), method 2 "extended fractal method with QCD integration": use of $N_c = 3$, α_s, Λ_{QCD} as external inputs.

Previous formulation

In the fermion tables of Doc. 006/046 each row carries a column "symmetry" with the following entries: **electron, muon, tau** Entry in "symmetry": lepton number.

ν_e, ν_μ, ν_τ Entry in "symmetry": double-ξ.

up, charm, strange, top, bottom Entry in "symmetry": color.

down Entry in "symmetry": color + isospin.

This mixture combines four different concepts in one column: an FFGFT scaling class ("double-ξ"), an SM conserved quantity ("lepton number"), an SM gauge symmetry ("color") and an SM quantum number ("isospin"). In addition, the entries "color charge" in Doc. 042 and $N_c = 3$ in Doc. 005 suggest that the SM color symmetry $SU(3)_C$ is an independent component of FFGFT.

Refined formulation

In FFGFT neither the gauge group $SU(3)_C$ nor a color quantum number exists. Quark masses follow, just like lepton masses, exclusively from the Yukawa resonance formula

$$m_f = r_f \cdot \xi^{p_f} \cdot v \tag{190.13}$$

with r_f a rational prefactor and p_f a rational exponent (step size $\Delta p = 1/3$). This is numerically substantiated by the script `calc-resonanz-leiter.py`: all six quarks hit their PDG masses to $\sim 0{,}3$–$3{,}4\,\%$, *without* use of N_c, α_s, Λ_{QCD} or a color quantum number.

Corrected column structure of the fermion table

The old column "symmetry" is replaced by two clearly separated columns:

electron, muon, tau FFGFT scaling class: single-ξ; SM correspondence (informative): lepton number.

ν_e, ν_μ, ν_τ FFGFT scaling class: double-ξ; SM correspondence (informative): lepton number, lepton mixing.

up, charm, top FFGFT scaling class: cluster-ξ; SM correspondence (informative): up-type, $T_3 = +1/2$.

down, strange, bottom FFGFT scaling class: cluster-ξ; SM correspondence (informative): down-type, $T_3 = -1/2$.

Meaning of the classes:

- *single-ξ*: $m = r\xi^p v$ with $p \in \{2/3, 1, 3/2\}$.
- *double-ξ*: $m_\nu = r_\nu \xi^2 m_e$ (additional ξ suppression relative to the charged leptons).
- *cluster-ξ*: $m = r\xi^p v$ with the same exponents $p \in \{-1/3, 1/2, 2/3, 1, 3/2\}$ as for the leptons, but with multi-node binding in the sense of Doc. 049 (cubic self-interaction $\lambda_s (\delta m_q)^3$).

Language convention for quarks

"color charge" (implies an SU(3)$_C$ quantum number) Use: multi-node binding.

"three colors red/green/blue" (SM gauge index) Use: triple-node cluster (the triplet is geometry, not an $SU(3)$ index).

"$N_c = 3$ colors" Use: $N_{\text{cluster}} = 3$ (geometric node count in the baryon).

"color confinement" Use: cluster binding (cubic self-interaction).

Status of Doc. 005, method 2

Doc. 005 contains two calculation methods:

- *Method 1: direct geometric method.* Yukawa form per Eq. (190.13); parameter-free; accuracy $\sim 1\,\%$ for all fermions. **This is the FFGFT method.**
- *Method 2: extended fractal method with QCD integration.* Uses N_c, α_s, Λ_{QCD}, lattice-QCD calibration and ML. **This is a phenomenological bridge construction** for linking to the lattice-QCD literature, not part of the parameter-free FFGFT.

In external communication method 1 is to be cited. Method 2 may be used if explicitly labelled as a "bridge to lattice QCD" or "ML-calibrated extension".

Justification

The designations "color", "color charge", "N_c" import a concept of standard quantum field theory ($SU(3)_C$ as a gauge symmetry, three colors, eight gluons, color confinement as a non-perturbative property). Doc. 049 states explicitly that FFGFT contains none of these elements: "what we call ‚color' becomes high-energy node-binding patterns" (Doc. 049, §3.7), with a single cubic interaction term $\lambda_s\,(\delta m_q)^3$ instead of eight gluon fields.

The script validation (`calc-resonanz-leiter.py`, `calc_De.py`) shows: all quark masses follow from the same Yukawa formula as the leptons. The column "symmetry" of the fermion table, by contrast, suggests that $SU(3)_C$ is a distinguishing property of the quarks within FFGFT – which is not the case.

The corrected column structure separates the FFGFT-internal (*scaling class*) from the informative SM link (*SM correspondence*), parallel to the separation of "recursive α running" and "renormalization group" in Refinement 7.

.13 Refinement 10: Logical status of $K = 245$ and the factor $1{,}314$ in Doc. 009

Affected document

Doc. 009 (origin of ξ), section "why $K = 245$ is fundamental", subsection "geometric meaning".

Misleading presentation

Doc. 009 lists the following derivation chain for $K = 245$:

- $\varphi^{12} = 321{,}996$
- correction by the fractal structure: $(1-\xi)^2 \approx 0{,}999733$
- result: $321{,}996 \times 0{,}999733 \approx 321{,}87$
- "scaling to the mass range": $321{,}87/1{,}314 \approx 245$

The factor $1{,}314$ appears without derivation or independent physical justification.

Correct logical order

$K = 245$ is *not* derived from φ^{12} and $1{,}314$. The correct logical order is the reverse:

$$K = \frac{R_{pe}}{(4/3)^{\kappa}} = \frac{1836{,}153}{(4/3)^7} \approx 245{,}097 \tag{190.14}$$

K is fixed by the observed proton-electron mass ratio R_{pe} and the exponent $\kappa = 7$. It is an *empirical anchor*, not a result from first principles. The factor $1{,}314$ is defined implicitly as $\varphi^{12}/245 \approx 1{,}314$ and does not constitute an independent derivation of K.

What remains non-trivial

The exponent $\kappa = 7$ is the *only integer* for which the complete e-p-μ triangle closes simultaneously:

κ	$K \times (4/3)^\kappa$	agreement with R_{pe}
6	1376,6	$-25\,\%$
7	1835,4	✓ $0{,}04\,\%$
8	2447,2	$+33\,\%$

This uniqueness is a real condition, not a tautology. $K = 245$ then follows from $\kappa = 7$ and the empirical R_{pe}.

Corrected formulation for external communication

- **Avoid:** "$K = 245$ is derived from first principles; no free parameters."
- **Use:** "An empirical anchor: $K = 245$, fixed by R_{pe} and $\kappa = 7$. From this anchor follow five mass ratios with $< 0{,}04\,\%$ accuracy. The φ^{12} observation is a numerological consistency note, not a derivation."

.14 Refinement 11: T_{CMB} correction factor and K_{frac} clarification

Affected documents

Doc. 025 (cosmology), Doc. 003 (foundations), Doc. 133 (fractal correction derivation).

Problem A: K_{frac} does not close the T_{CMB} gap

Doc. 025 derives:

$$T_{\text{CMB}} = \frac{16}{9}\,\xi = \frac{4}{16875}\ \text{eV} \approx 2{,}7507\ \text{K} \tag{190.15}$$

The measured value is $T_{\text{CMB}} = 2{,}72548$ K (Planck 2018). The gap is $\approx 0{,}925\,\%$.

$K_{\text{frac}} = 1 - 100\xi = 74/75$ (Doc. 133) does *not* close this gap. Numerically:

$$\frac{16}{9}\,\xi \times K_{\text{frac}} = 2{,}714\ \text{K} \quad (0{,}42\,\%\ \text{below the target value, worse}) \tag{190.16}$$

The required correction factor is $0{,}99083$, achieved by the following formula:

$$\boxed{T_{\text{CMB}} = \frac{16}{9}\,\xi\left(1 - \frac{275}{4}\,\xi\right) = 2{,}725486\ \text{K}} \tag{190.17}$$

Agreement with the measured value: $\Delta = 0{,}0002\,\%$.

The coefficient $275/4 = 68{,}75$ lies close to the tetrahedral number $68 = 72 - 4$ (Doc. 003, section 0.5), but its derivation as a second-order correction is not yet formally established. **This is an open point:** a geometric derivation of $275/4$ is required. The formula above is a numerically verified result that still requires a complete justification.

Problem B: two inconsistent K_{frac} definitions

Doc. 003 gives:

$$K_{\text{frac}}^{(003)} = 1 - \frac{D_{\text{frac}} - 2}{68} = 1 - \frac{0{,}94}{68} \approx 0{,}98530 \tag{190.18}$$

where $D_{\text{frac}} = 2{,}94$ is given without derivation from ξ.

Doc. 133 gives:

$$K_{\text{frac}}^{(133)} = 1 - 100\xi = \frac{74}{75} \approx 0{,}98667 \tag{190.19}$$

derived from the cumulative RG running with $D_f = 3 - \xi$.

These values differ by $1{,}37 \times 10^{-3}$ and cannot both be correct. The **primary definition** is Doc. 133: $K_{\text{frac}} = 1 - 100\xi = 74/75$. The value $D_{\text{frac}} = 2{,}94$ in Doc. 003 is an independent approximation not derived from ξ; it uses a different (tetrahedral) model. Doc. 003 is to be understood as a heuristic motivation, not as a defining formula.

Corrected formulation for T_{CMB}

- **Avoid:** "K_{frac} closes the T_{CMB} gap."
- **Use:** "T_{CMB} requires a second-order correction $(1 - 275/4 \cdot \xi)$ that yields $0{,}0002\,\%$ agreement. The geometric origin of $275/4$ is still open."

Addendum: ΔCHSH $\approx 10^{-5}$ in Doc. 230

Doc. 230 claims ΔCHSH $\approx 10^{-5}$ as a geometrically motivated prediction. Doc. 022 gives the formula

$$\text{CHSH}(N) = 2\sqrt{2} \cdot \exp\left(-\xi \frac{\ln N}{D_f}\right) \tag{190.20}$$

where N is the number of measurement runs. This formula yields ΔCHSH $\approx 5 \times 10^{-4}$ at $N = 73$ (not 10^{-5}). ΔCHSH grows monotonically with N; at $N = 2$ (minimum) one already obtains ΔCHSH $\approx 8{,}7 \times 10^{-5}$. The value 10^{-5} from Doc. 230 is not attainable at any finite N under the Doc. 022 formula.

Important distinction (Onur Teker, May 2026): CHSH$(N) \to 0$ (correlation vanishes) requires $N \to \infty$ and is a different condition from ΔCHSH $= 10^{-5}$ (distance from the Tsirelson bound). Both conditions confirm that the Doc. 022 formula cannot reproduce the Doc. 230 value. The prediction ΔCHSH $\approx 10^{-5}$ requires a separate derivation or a suppression mechanism that is not yet documented. **Open point.**

.15 Refinement 12: Two different ΔCHSH observables in Doc. 022 and Doc. 230

Affected documents

Doc. 022 (QFT-ML addendum), Doc. 230 (Hilbert-space translation).

The apparent contradiction

Doc. 022 gives ΔCHSH $\approx 5 \times 10^{-4}$ at $N = 73$ via the formula CHSH$(N) = 2\sqrt{2} \cdot \exp\left(-\xi \ln N / D_f\right)$. Doc. 230 claims ΔCHSH $\approx 10^{-5}$ without derivation. These values differ by a factor of ~ 50.

Possible resolution: cumulative vs. elementary deviation

The two values refer to different observables:

- **Doc. 022** describes the *cumulative* deviation over N measurement runs. It grows monotonically with N: $\Delta\text{CHSH}(N) \sim \xi \ln(N)/D_f \times 2\sqrt{2}$. This is the experimentally relevant quantity for a Bell test with N runs.
- **Doc. 230** plausibly refers to the *elementary* deviation per single measurement – the phase cost of one θ_4 projection cycle. The natural candidate is:

$$\Delta\text{CHSH}_{\text{elem}} = \frac{\xi}{2\pi} \approx 2{,}1 \times 10^{-5} \approx 10^{-5} \tag{190.21}$$

These are the Landauer costs per measurement expressed as a CHSH deviation: the θ_4 phase runs through a full 2π cycle per measurement, and each cycle costs ξ in geometric units.

The ratio between cumulative and elementary deviation at N runs:

$$\frac{\Delta\text{CHSH}(N)}{\Delta\text{CHSH}_{\text{elem}}} \approx N \cdot \frac{2\pi}{D_f} \tag{190.22}$$

At $N = 73$ this gives a factor of ≈ 153, consistent with the observed factor of ~ 50–100 between the two values.

Status

This resolution is a **working hypothesis**, not a proven derivation. The identification $\Delta\text{CHSH}_{\text{elem}} = \xi/(2\pi)$ requires a formal justification from the T⁴ projection geometry. If confirmed, there is no contradiction between Doc. 022 and Doc. 230 – they describe the same physical effect at different integration scales.

Experimental consequence: The cumulative deviation $\Delta\text{CHSH}(N) \approx 5 \times 10^{-4}$ at $N = 73$ is within reach of future loophole-free Bell tests (current resolution $\sim 10^{-2}$; required improvement: factor ~ 20). The elementary deviation $\xi/(2\pi) \approx 2 \times 10^{-5}$ would require single-shot precision beyond current technology.

.16 Refinement 13: Bridge formula between ξ_{FFGFT} and ξ_{UIFT}

Context

In the IPI discussion (May 2026) Onur Teker (UIFT) writes:

> $\xi = k_B T_{\text{CMB}} \ln 2/(m_e c^2)$, derived from three independent measurements (T_{CMB}, k_B, m_e). This is not a hyperparameter – it is a prediction.

This identifies ξ_{UIFT} with $\xi_{\text{FFGFT}} = 4/30000$. Numerically they differ considerably. This refinement documents the exact relation.

The key: FFGFT uses eV as the temperature unit

In FFGFT the natural unit system $\hbar = c = k_B = 1$ holds. Temperatures are expressed in eV (not in Kelvin):

$$T_{\text{CMB}}[\text{eV}] = k_B \cdot T_{\text{CMB}}[\text{K}] = k_B \cdot 2{,}72548\,\text{K} = 2{,}3486 \times 10^{-4}\,\text{eV} \tag{190.23}$$

The FFGFT formula (Doc. 025) then reads – to first order:

$$T_{\text{CMB}}[\text{eV}] = \frac{16}{9}\,\xi_{\text{FFGFT}} = 2{,}3704 \times 10^{-4}\,\text{eV} \qquad (\Delta = 0{,}93\,\%) \tag{190.24}$$

and to second order (Doc. 190 P11):

$$T_{\text{CMB}}[\text{eV}] = \frac{16}{9}\,\xi_{\text{FFGFT}}\left(1 - \frac{275}{4}\,\xi_{\text{FFGFT}}\right) = 2{,}3486 \times 10^{-4}\,\text{eV} \qquad (\Delta = 0{,}0002\,\%) \tag{190.25}$$

Bridge formula

Inserting (190.24) into Onur's expression (first order):

$$\xi_{\text{UIFT}} = \frac{k_B T_{\text{CMB}}[\text{K}]\ \ln 2}{m_e c^2} = \frac{T_{\text{CMB}}[\text{eV}] \cdot \ln 2}{m_e c^2/\text{eV}} = \frac{\frac{16}{9}\,\xi_{\text{FFGFT}} \cdot \ln 2}{m_e c^2/\text{eV}} \tag{190.26}$$

Rearranged:

$$\boxed{\frac{\xi_{\text{FFGFT}}}{\xi_{\text{UIFT}}} = \frac{m_e c^2/\text{eV}}{\frac{16}{9} \cdot \ln 2} = \frac{510\,999\,\text{eV}}{1{,}2323\,\text{eV}} \approx 414\,684} \tag{190.27}$$

SI conversion table (reconstructed from Doc. 013, archived version)

The following table was documented in older versions of Doc. 013 (SI conversions for the natural FFGFT unit system, since shortened). It is reproduced here because it is essential for the bridge formula.

Quantity	Nat. units	SI / eV
temperature unit $[T]$	$= [E]$	1 eV $=$ $k_B^{-1} \cdot 1$ eV $=$ $11{,}604$ K
T_{CMB} [nat.]	$(16/9) \cdot \xi = 2{,}370 \times 10^{-4}$	$2{,}370 \times 10^{-4}$ eV
T_{CMB} [K] (Planck 2018)	—	$2{,}72548$ K
T_{CMB} [eV] $= k_B \cdot T$[K]	—	$2{,}3486 \times 10^{-4}$ eV
$m_e c^2$	$= m_e$ (nat.)	$510{,}999$ eV $= 0{,}511$ MeV
k_B	$= 1$ (nat.)	$8{,}617 \times 10^{-5}$ eV/K
$\hbar$	$= 1$ (nat.)	$6{,}582 \times 10^{-16}$ eV·s
c	$= 1$ (nat.)	$2{,}998 \times 10^{8}$ m/s

Numerical verification

Quantity	Value	Unit
$\xi_{\text{FFGFT}} = 4/30000$	$1{,}3333 \times 10^{-4}$	dimensionless
$\xi_{\text{UIFT}} = k_B T_{\text{CMB}} \ln 2/(m_e c^2)$	$3{,}1858 \times 10^{-10}$	dimensionless
ratio $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}}$	$418\,521$	—
scale factor $m_e c^2/(\frac{16}{9} \ln 2)$ [eV]	$414\,684$	—
T_{CMB} [K] (Planck 2018)	$2{,}72548$	K
T_{CMB} [eV] $= k_B \cdot T_{\text{CMB}}$[K]	$2{,}3486 \times 10^{-4}$	eV
$(16/9) \cdot \xi$ [eV] (FFGFT 1st order)	$2{,}3704 \times 10^{-4}$	eV
$(16/9) \cdot \xi(1 - 275/4 \cdot \xi)$ [eV] (P11)	$2{,}3486 \times 10^{-4}$	eV
$m_e c^2$	$510\,999$	eV

The residual deviation of $0{,}93\,\%$ between the ratio ($418\,521$) and the scale factor ($414\,684$) is exactly the first-order T_CMB error – this confirms the correctness of the bridge formula.

Physical interpretation

ξ_{FFGFT} and ξ_{UIFT} are **not the same quantity**. They are connected by a pure SI conversion factor: the ratio of the relativistic electron energy ($m_e c^2 = 511$ keV) to the eV scale of the FFGFT temperature formula, modulated by $(16/9) \cdot \ln 2$.

The SI conversion table above was present in older versions of Doc. 013 (archived March 2026, Git commit `f2be9c8`) and has since been shortened. The explicit statement $T_{\text{CMB}}^{\text{nat}} = k_B \times 2{,}725$ K $= 2{,}35 \times 10^{-4}$ eV is found verbatim in that version. The reconstruction here restores the connection between the natural-unit formula and the SI value, which was implicitly assumed but no longer shown.

The Vopson-Teker experiment measures ξ_{UIFT} directly. If confirmed, that evidences the thermodynamic ground state at T_{CMB} – not $\xi_{\text{FFGFT}} = 4/30000$. The geometric derivation of ξ_{FFGFT} from $\kappa = 7$ (Doc. 009) is a separate and independent condition.

.17 Refinement 14a: Redshift in Doc. 041 – mechanism instead of energy loss

Affected document

Doc. 041 (parameter derivation), table "Cosmological Parameters", LEVEL 3 (redshift) – in particular the description "photon energy loss through ξ-field".

Previous formulation

The cosmological redshift is labelled in the comparison table as "photon energy loss through ξ-field" – i.e. as an energy loss of the photon in the ξ field.

Refined formulation

The physical mechanism is the **fractal path lengthening**: the geometric stretching of the propagation path in the ξ field (torsion lattice), as derived authoritatively in Doc. 026, Doc. 149 and Doc. 182. The dependence on the path length d (and the wavelength) remains and is consistent with the fractal path lengthening – it is *not* a contradiction.

An "energy loss" of the photon is **not an independent mechanism** but an *apparent artifact*: it would arise only if one erroneously fails to assume the geometric path lengthening. The wording "photon energy loss" in Doc. 041 is therefore to be replaced by "fractal path lengthening (geometric path stretching)"; the "energy loss" may at most be named explicitly as the artifact that would arise with the path lengthening omitted.

Justification

The static FFGFT cosmology explains the redshift purely geometrically (Doc. 182: "not by an intrinsic energy loss at the photon, but by geometric stretching of the propagation path"; *tired light* does not occur). The comparison table in Doc. 041 abbreviates this mechanism to "energy loss" and thereby evokes precisely the tired-light connotation that FFGFT does not intend. The refinement brings Doc. 041 into line with the authoritative derivation (Doc. 026/149/182) without changing the path-length dependence.

Status of incorporation

Incorporated (as of 2 June 2026). Implemented in Doc. 041 (DE+EN).

.18 Refinement 15: Mark cosmological z as a ΛCDM comparison quantity

Affected documents

Doc. 061 (temperature units/CMB), section "modified recombination history" as well as abstract/summary; Doc. 041 (comparison tables ΛCDM vs. T0).

Previous formulation

A z-dependent recombination calculation ($z_{\text{rec}}^{T0} \approx 1089{,}5$, $T(z) = T_0(1+z)[\dots]$) stood as "In T0 theory recombination occurs at ..." – i.e. as an FFGFT-internal quantity. Likewise "CMB at $z \approx 1100$" in the abstract/summary.

Refined formulation

The static ξ universe has *no* cosmological z in the expansion sense; FFGFT's redshift is fractal path lengthening, not a scale factor. Therefore z in these calculations is a **ΛCDM pipeline quantity** and is to be marked explicitly as a *comparison calculation*, not as an FFGFT result. In FFGFT the CMB arises through stationary thermalization in an infinitely old universe, not through decoupling at some z. In Doc. 061 the section is framed accordingly; the abstract/summary places are changed to "without decoupling at any z". In Doc. 041 the $z = 1100$ values are marked as "(ΛCDM)" (they stand in the ΛCDM comparison column anyway).

Justification

Unmarked comparison calculations create the impression of FFGFT-internal results and violate the standing caution that cosmological pipeline quantities (H_0, z, Λ_{obs}) must not be presented as FFGFT statements without labeling. The marking restores the separation without removing the comparison numbers.

Status of incorporation

Incorporated (as of 2 June 2026). Docs. 061 and 041 (DE+EN) marked. Further unmarked pipeline comparisons in the corpus are to be checked separately (ongoing).

.19 Refinement 16: H_0 is emergent (decompactified), not fundamental

Affected documents

Doc. 026 (geometric cosmology), l. 116/117 DE+EN; with offshoots in Doc. 064 (l. 104) and Doc. 181 (l. 158). *Already correctly implemented in Doc. 263* (H_0 as an emergent, decompactified quantity); the remaining places are here only *noted*, not yet changed.

Previous formulation (outdated)

Doc. 026 calls H_0 a "fundamental constant describing the interaction of light with the geometry of the T0 vacuum". This formulation stems from an earlier state that still understood the redshift as an *energy loss* of light passing through the vacuum (a near-"tired light" picture). Doc. 064 ("measures energy loss in the ξ-field") and Doc. 181 ("geometric energy loss") carry the same remnant.

Refined view

Two independent reasons argue against treating H_0 as a fundamental constant:

- **Energy loss is outdated (cf. P14a):** The redshift is fractal path lengthening, not energy loss. The "vacuum as a traversed medium" is itself already a decompactified description and not part of the compactified T^4 view.
- H_0 **is emergent, not fundamental:** Doc. 016 records that age, critical density and Hubble length are *Friedmann concepts without a direct T0 counterpart*; Doc. 262, that $\hbar, \varepsilon_0$ become *real only upon decompactification*. Since $H_0 = E_H/\hbar$ contains the (only then real) $\hbar$, H_0 does not exist in the compactified T^4 form. H_0 is an emergent quantity of the decompactified description; the non-integer exponent $41/4$ is an expression of this scale transition, not a fundamental geometry constant. (The integer ansatz $n = 10$ in Doc. 165 is therefore to be viewed with caution – it forces an emergent transition quantity into a compact geometry.)

Status of incorporation

Partially incorporated (as of 2 June 2026). Implemented in Doc. 263 (H_0 emergent/decompactified). **Open:** Doc. 026 (DE+EN, l. 116/117) is a document that is *outdated* in content (energy-loss/vacuum picture) and requires a more fundamental revision or a labeling as historical; likewise the offshoots in Doc. 064/181. Noted as a larger construction site.

.20 Refinement 17: circular/fit-dependent DE-DM relations in Doc. 028

Affected document

Doc. 028 ("7 questions"), sections "Riddle 4: MOND" and "Riddle 5: Dark Energy and Dark Matter". Only *noted*, not yet changed.

Finding (recomputed)

- **DE/DM ratio: circular as a *prediction*, admissible as a *conversion*.** Doc. 028 gives $\rho_{\text{DE}}/\rho_{\text{DM}} = \xi^{\alpha}$ with $\alpha = \ln(2{,}5)/\ln(\xi)$. Inserting this α into ξ^{α} yields *exactly* $2{,}5$ *for any* ξ (checked with $\xi = 10^{-3}, 10^{-5}, 0{,}5$). *As a prediction* from the ξ geometry the relation is thus circular (violates Doc. 101). *But:* the target value $2{,}5$ ($\Omega_{\Lambda}/\Omega_{\text{DM}}$) is itself not a model-neutral measured value but a ΛCDM pipeline output (standing caution). Reading the relation therefore *not* as a prediction but as a *conversion factor* between FFGFT geometry and the ΛCDM-Ω bookkeeping – analogous to c (meter/second) and to z (cf. P15) – the circular calibration via $2{,}5$ is *admissible*. **Limitation:** unlike c (both sides have the same real counterpart), the ΛCDM side with ρ_{DE} contains a model artifact (Λ does not exist in FFGFT). The factor thus translates partly a real quantity (the DM bookkeeping $\leftrightarrow$ ξ effect), partly an artifact (ρ_{DE}). Therefore: admissible, but *only of limited* informativeness – as far as the ΛCDM quantity is artifact-free.
- **The MOND scale is fit-dependent.** $a_0/(cH_0) = \xi^{1/4} \cdot K_M$ gives the measured value $0{,}176$ only with $K_M = 1{,}637$. The $\xi^{1/4} = 0{,}1075$ is genuinely from ξ; K_M is a *free adjustment factor* without derivation. Partially derived, not parameter-free.
- **Two incompatible a_0 formulas; rotation curve open.** Besides Doc. 028, Doc. 201 (§4.3) gives a *different* form: $a_0 = c^2\xi/(4\lambda) \approx 1{,}2\times10^{-10}$ m/s^2. This is *not reproducible*: for the correct unit (m/s^2) λ would have to be a *length* ($\sim 2{,}5\times10^{22}$ m), whereas in the rest of Doc. 201 λ is via $m_T = \lambda/\xi$ an *energy/mass* – the same symbol in two incompatible meanings. Without an independent λ value, a_0 is either undetermined or (backward from the measured value) circular; the length scale $2{,}5\times10^{22}$ m corresponds to no known FFGFT scale. *Consequence:* both a_0 formulas (028: fit-dependent; 201: dimensionally inconsistent) do not carry. The *flat rotation curve* follows qualitatively as a MOND consequence of the ξ geometry (static, without DM substance; Doc. 201/145/156), but the concrete profile $v(r)$ and the SPARC comparison are *not worked out* – remains open.

Delimitation (what remains valid)

The CMB relations are *not* affected and remain robust: $T_{\text{CMB}} = \frac{16}{9}\xi^2 E_\xi$ and $|\rho_{\text{Casimir}}|/\rho_{\text{CMB}} = \pi^2/(240\xi) \approx 308$ contain ξ alone, without a subsequently adjusted exponent or fit factor.

Status of incorporation

Noted (as of 2 June 2026), not yet changed. Recommendation: the DE/DM ratio (Riddle 5) is to be marked *not* as a prediction but as a limited conversion factor to the ΛCDM-Ω bookkeeping (admissible like c/z, but bounded by the ρ_{DE} artifact); the MOND scale (Riddle 4) as partially derived with a free K_M. The *interpretation* of the dark sector is settled – DM is a ξ-geometric effect (an artifact of the ΛCDM view, like H_0 and z), DE is eliminated (no Λ). *Quantitatively* it remains open: no model-neutrally derived Ω_{DM}, no worked-out rotation curve.

.21 Refinement 14b: CMB anharmonicity, forward vs. retrodiction, and the $|n|^2 = 30$ case (P29/P30/P31)

An explicit forward investigation (Doc. 268, step 17; scripts `python/Dok267_268_Skripte/`) clarified the status of the CMB peak sector and split it into three points.

(1) The direction of the inference (P29). The earlier presentation created the impression that $\{1, 6, 14, 26\}$ and hence $\{3, 18, 42, 78\}$ follow forward from the T^4 geometry. In fact $\{1, 6, 14, 26\}$ is obtained by squaring the *measured* CMB ratios (Doc. 268, step 4). It is a **retrodiction** (consistency check), not a parameter-free prediction. The genuine forward maxima of the full T^4 spectrum are dense ($|n|^2 = 3, 6, 10, 14, 18, 22, 26, 30, \ldots$).

(2) The $|n|^2 = 30$ case (P29, sharpened by O. Teker). $30 = 3 \cdot 10$ satisfies the factor-3 filter, since $|n|^2 = 10$ is a local maximum. Thermally one even has

$$I(30) = 13{,}2 \ > \ I(42) = 7{,}6 \ > \ I(78) = 1{,}7,$$

so 30 ($\ell \approx 696$) would be *stronger* than two visible peaks, yet shows no peak in the CMB (it lies in the trough between peak 2 and 3). The earlier argument "strongest local maximum in the range" fails here. The selection mechanism is thus *not* provided by the factor 3 alone. *Open.*

(3) The naive C_ℓ Bessel projection is insufficient (P30). The calculation $C_\ell = \sum_n N_{\mathrm{BE}} \, |j_\ell(k_{\mathrm{obs}} D_A)|^2$ does *not* reproduce $\{3, 18, 42, 78\}$; it washes out to a quasi-continuum dominated by the lowest modes. The earlier P30 formulation ("a feasible calculation, no fundamental obstacle") is withdrawn: a physical source/window function is missing. *Open.*

(4) The forward content – geometric anharmonicity (new, P31). Forward, the symmetric T^4 resonance generates the *harmonic* backbone $1 : 2 : 3 : 4 : 5$ ($k^2_{\mathrm{obs}} = 3n^2$). The CMB is *anharmonic* ($1 : 2{,}44 : 3{,}68 : 5{,}09$). This anharmonicity follows forward and parameter-free from the codimension-1 projection:

$$D_{\mathrm{eff}} = D_f - 1 = 2 - \xi \approx 2 \ \Rightarrow \ \nu = \frac{D_{\mathrm{eff}}}{2} - 1 = -\frac{\xi}{2} \approx 0 \ \Rightarrow \ J_0 \text{ membrane} \ \Rightarrow \ 1 : 2{,}30 : 3{,}60 : 4{,}90,$$

matching to $\sim 5\%$, *without* mode selection. The radial Bessel index is $\nu = D/2 - 1$; the anharmonicity is maximal at $D = 2$ and vanishes at $D = 1$ and $D = 3$. The CMB sits at the dimension of maximal anharmonicity. The match is to the *flat* membrane (J_0), not to the curved 2-sphere ($Y_{\ell m}$, gives $1 : 1{,}73 : \ldots$, wrong). This is the conceptually clean, forward-directed statement; it removes the circularity objection because no selection from data is needed.

(5) The residual deviation – not from ξ. The best fit lies at $D \approx 1{,}86$ (just below 2). A spectral dimension $d_s < D_f$ would be the obvious source but fails: the FFGFT fractal is too weakly fractal ($d_w = 2 + O(\xi) \Rightarrow d_s \approx 2$); for $d_s = 1{,}86$ one would need $d_w \approx 2{,}15$ ($\sim 1000 \times \xi$). The deviation is not derivable from ξ and is of the order of the measurement uncertainty (J_0 is consistent with peaks 3,4 within $\sim 2\sigma$; only peak 2 at $\sim 4\sigma$). Pointing to an exact value would be number-juggling. *Open, presumably partly measurement noise.*

Consequence. The defensible core reads: harmonic T^4 backbone (forward) + geometric anharmonicity from the codimension-1 projection (forward, $\sim 5\%$). The ratio match $\{3, 18, 42, 78\}$ is a retrodiction; the rigorous single-peak selector and the exclusion of $|n|^2 = 30$ remain open. The particle sector (α, lepton masses, Koide) is *unaffected* by this – it rests on direct ξ powers, not on mode counting.

.22 Register of corrections, clarifications and revisions

The following entries are sequential and append-only: an entry once booked is not overwritten but supplemented, clarified or withdrawn by a later one. The source documents are not revised (cf. R50).

C1 — Doc. 049

Concerns: 049 (HTML)

Erroneous / unclear. $\xi = \lambda_h^2 v^2/(64\pi^4 m_h^2)$

Correct / clarified. $\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$ *[Addendum 28 July 2026, cf. R66: reason for the correction —* $16\pi^3 = (4\pi)\cdot(4\pi^2) = S^2 \cdot 2S^3$ *is the sphere count of the compact* T^4 *boundary geometry (Doc. 310); the earlier version counted one 2-sphere too many]*

C2 — Doc. 116

Concerns: 116 (tab.)

Erroneous / unclear. $r_\tau = 8/3$

Correct / clarified. $r_\tau = 25/9$

C3 — Doc. 018

Concerns: 018 (Explorer)

Erroneous / unclear. $a_e = 4\pi/(f\cdot k)$

Correct / clarified. $a_e = 2\pi^2/(f\cdot k)$

C4 — Doc. 267

Concerns: 267 (DE+EN), 268 (DE)

Erroneous / unclear. factor 3 interpreted as the fractal dimension D_f ("fractal dispersion relation $k^2_{\text{eff}} = D_f k^2$", Doc. 051)

Correct / clarified. Factor 3 = three observable space dimensions: the observer measures $k^2_{\text{obs}} = (n_1^2 + n_2^2 + n_3^2)/L^2$ (the fourth torus direction is time unfolding). The conformal metric (Doc. 051) cancels for massless particles; the fractal correction (Doc. 204) is $+0{,}69\,\xi k^2 \approx 10^{-5}$, *not* the factor 3. $D_f = 3 - \xi$ is consistent with factor 3 but is not its origin. *Done (267 DE+EN, 268 DE), June 2026*

C5 — Doc. 275

Concerns: 275 (script v3)

Erroneous / unclear. Three implementation errors in `ffgft_hlv_gap_v3.py`: (a) the T^4 point lattice used n_3 only in the acceptance norm, not in the position (729 unique out of 2000 points; the $1/d^2$ clamp produced $\lambda_{\text{max}} = 2{,}5\cdot 10^{12}$) – the result "gap_CV $= 1{,}68$ outside the band" was an artifact; (b) `hlv_points(seed)` did not use the seed – the 3-seed "ensemble" was the same calculation three times; (c) the shell detector for target ratios < 1 was structurally biased (minimum error $(1{,}05/c - 1)$; a larger sub-1 ratio wins on every point set) – the discrimination $\Delta = 0{,}145$ was structural

Correct / clarified. **All three fixed in v4** (11 June 2026): T^4 as kNN in genuine 4D coordinates with dimension-matched 4D nulls (result: $1{,}329 \pm 0{,}025$ *inside* the band $[0{,}690,\ 2{,}406]$); HLV ensemble with $\sigma = 0{,}05$ ($0{,}710 \pm 0{,}008$, in the band); the shell detector raises an exception for targets ≤ 1. v3 results in Doc. 275 visibly retracted. *Audit: P. Stoychev (IPI), 11 June 2026*

P1 — Doc. 116 / 158

Concerns: 116 / 158
Erroneous / unclear. $\Delta Q < 0{,}00003\,\%$ vs. $0{,}001\,\%$
Correct / clarified. both correct; $0{,}001\,\%$ for external use

P2 — Doc. several

Concerns: several
Erroneous / unclear. $\hbar$ as a postulate
Correct / clarified. $\hbar$ follows from ξ via $L_0 = \xi \cdot \ell_P$

P3 — Doc. 173

Concerns: 173–176
Erroneous / unclear. a single uniform ξ value
Correct / clarified. two parameters: ξ_0 (geom.) and ξ_{Higgs} (alg.)

P4 — Doc. 180

Concerns: 180–182
Erroneous / unclear. L_0 as a Schwarzschild radius
Correct / clarified. L_0 = geom. fundamental length

P5 — Doc. several

Concerns: several
Erroneous / unclear. "FFGFT computes masses exactly"
Correct / clarified. mass structure from ξ; $\sim 1\,\%$ agreement

P6 — Doc. 116 / 158

Concerns: 116 / 158
Erroneous / unclear. Koide with FFGFT symbolic expressions
Correct / clarified. Koide only with numerical values (non-closure, Doc. 193)

P7 — Doc. 005, 008, 019, 086, 189, 202

Concerns: 005, 008, 019, 086, 189, 202
Erroneous / unclear. "renormalization group" as an FFGFT tool
Correct / clarified. scale structure from the recursion $\mathscr{R}$

P8 — Doc. 005, 008, 012, 014, 041, 060, 133, 141, 159, 181

Concerns: 005, 008, 012, 014, 041, 060, 133, 141, 159, 181
Erroneous / unclear. "fractal renormalization" as an internal term
Correct / clarified. geometric scale adjustment / fractal correction

P9 — Doc. 005, 006, 042, 046

Concerns: 005, 006, 042, 046
Erroneous / unclear. column "symmetry" with "color";
$N_c = 3$ as an external SM input
Correct / clarified. columns "scaling class" + "SM correspondence";
quarks via $m = r\xi^p v$

P10 — Doc. 009

Concerns: 009
Erroneous / unclear. $K = 245$ from $\varphi^{12}/1{,}314$;
"no free parameters"
Correct / clarified. $K = R_{pe}/(4/3)^7$: an empirical anchor;
factor $1{,}314$ defined implicitly, not derived

P11 — Doc. 025, 003, 133

Concerns: 025, 003, 133
Erroneous / unclear. K_{frac} closes the T_{CMB} gap;
one definition of K_{frac}
Correct / clarified. correct: $(1 - 275/4 \cdot \xi)$, $\Delta = 0{,}0002\,\%$;
Doc. 003 and Doc. 133 use different models

P12 — Doc. 022, 230

Concerns: 022, 230
Erroneous / unclear. ΔCHSH $\approx 10^{-5}$ (Doc. 230) vs. 5×10^{-4} at $N = 73$ (Doc. 022)
Correct / clarified. two different observables: cumulative (Doc. 022) vs. elementary $\xi/(2\pi)$ (Doc. 230); 10^{-5} unreachable at any finite N from Doc. 022

P13 — Doc. 013, 025, 061

Concerns: 013, 025, 061
Erroneous / unclear. $\xi_{\text{FFGFT}} = \xi_{\text{UIFT}}$;
SI conversion table missing
Correct / clarified. $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} = m_e c^2/(\frac{16}{9}\ln 2 \cdot \text{eV}) \approx 414\,684$;
table in Doc. 013 (archived) reconstructed

P14a — Doc. 041

Concerns: 041

Erroneous / unclear. redshift as "photon energy loss through ξ-field"

Correct / clarified. fractal path lengthening (geometric path stretching); energy loss only an artifact when the path lengthening is omitted; path-length dependence remains (Doc. 026/149/182); the same path lengthening carries the CMB treatment in Doc. 268 (formerly tracked separately as P19)

P15 — Doc. 061 / 041

Concerns: 061 / 041

Erroneous / unclear. cosmological z ($z \approx 1100$, $z_{\text{rec}} \approx 1089{,}5$) unmarked as an FFGFT quantity

Correct / clarified. marked as a ΛCDM comparison quantity; the static universe has no cosmological z; CMB by stationary thermalization

P16 — Doc. 026 / 064 / 181

Concerns: 026 / 064 / 181

Erroneous / unclear. H_0 as a "fundamental constant" (energy-loss/vacuum picture)

Correct / clarified. H_0 is emergent (decompactified), not fundamental; implemented in Doc. 263, Docs. 026/064/181 noted as outdated

P17 — Doc. 028

Concerns: 028

Erroneous / unclear. DE/DM ratio $\xi^{\ln(2{,}5)/\ln\xi}$ as a prediction; MOND with fit factor K_M

Correct / clarified. the DE/DM relation is circular (gives 2.5 for any ξ); MOND partially derived ($\xi^{1/4}$ genuine, K_M free); CMB relations unaffected; noted

P18 — Doc. 267

Concerns: 267

Erroneous / unclear. the relation ΛCDM expansion vs. FFGFT time unfolding treated as an empirical question

Correct / clarified. the two pictures are *not* distinguishable through the observed ratios; the decision is theoretical, not empirical (conceptual placement)

P20 — Doc. 267 / 268

Concerns: 267 / 268

Erroneous / unclear. absolute cosmological scale (R_H, D_A, z_*) presented as an FFGFT result

Correct / clarified. the *form* is fixed: a dimensionless ξ ratio (R_H/ℓ_P). The *factor* (exponent $41/4$, $E_H = E_0\,\xi^{41/4}$, Doc. 182) is *not* derived from the T^4 geometry and must not be fitted (fitting = retrodiction). FFGFT is not the traditional yardstick; the factor is therefore adopted as a *declared external calibration* from ΛCDM – a choice of units, not an adoption of physics – and inserted with the converged value once the H_0 tension (~67 vs. ~73) is resolved; parametrized until then. The imported value does *not* count as FFGFT confirmation. Affects only absolute positions ($\ell_1 \approx 220$, $z_* \approx 875$), not the ratios; the internal forward derivation of the factor remains the goal. *Open*

P29 — Doc. 268

Concerns: 268

Erroneous / unclear. selection rule: why does the CMB show only the subset $\{3, 18, 42, 78\}$ of the T^4 peaks?

Correct / clarified. *Only partially answered.* The factor 3 from the 3D projection ($3 \cdot \{1, 6, 14, 26\}$) is plausible, but the sequence $\{1, 6, 14, 26\}$ is *read off the data* (Doc. 268, step 4), not produced forward. In particular $|n|^2 = 30 = 3 \cdot 10$ is *not* excluded: $|n|^2 = 10$ is a local maximum, and thermally $I(30) = 13{,}2 > I(42) = 7{,}6 > I(78) = 1{,}7$ – so 30 ($\ell \approx 696$) should be a peak, yet sits in a CMB trough. The "strongest in range" argument fails. *Open*, sharpened by O. Teker

P30 — Doc. 268

Concerns: 268

Erroneous / unclear. rigorous justification of the selection mechanism (full C_ℓ projection) missing

Correct / clarified. the *naive* spherical Bessel sum $C_\ell = \sum_n N_{\mathrm{BE}} |j_\ell(k_{\mathrm{obs}} D_A)|^2$ does *not* reproduce $\{3, 18, 42, 78\}$ – it washes out to a quasi-continuum dominated by low modes. The earlier formulation "a feasible calculation, no fundamental obstacle" was too optimistic: a physical source/window function selecting the peaks is missing. *Open*

P31 — Doc. 268

Concerns: 268

Erroneous / unclear. anharmonicity of the CMB peaks: whence the stretch relative to the harmonic $1 : 2 : 3 : 4 : 5$?

Correct / clarified. *Leading behaviour derived forward.* Forward, T^4 gives the harmonic backbone $1 : 2 : 3 : 4 : 5$ (symmetric resonance $k^2 = 3n^2$). The observed anharmonicity $1 : 2{,}44 : 3{,}68 : 5{,}09$ follows from the codimension-1 projection: last-scattering surface $\Rightarrow D_{\mathrm{eff}} = D_f - 1 = 2 - \xi \approx 2 \Rightarrow$ Bessel index $\nu = -\xi/2 \approx 0 \Rightarrow J_0$ membrane $\Rightarrow 1 : 2{,}30 : 3{,}60 : 4{,}90$, matching to $\sim 5\%$, parameter-free, without mode selection. The small residual ($D \approx 1{,}86$) is *not* derivable from ξ (the spectral dimension of the weakly fractal space stays ≈ 2) and is of the order of the measurement uncertainty. *Partially derived*; remainder open

P32 — Doc. corpus-wide

Concerns: corpus-wide

Erroneous / unclear. statements presenting H_0 or the absolute scale as *derived by FFGFT*

Correct / clarified. follows from P20: H_0 is an *imported* external calibration, not derived. All passages presenting H_0 as an FFGFT result are to be cleaned up ("imported/declared" instead of "derived"). *Noted for now*; corpus-wide cleanup deferred. *Open*

P33 — Doc. 009 / 042 / 181

Concerns: 009 / 042 / 181

Erroneous / unclear. magnitude factor 5^4 in $\xi = 1/(3 \cdot 2^2 \cdot 5^4)$ presented as *derived*

Correct / clarified. the *form* of ξ is forward: the harmonic $4/3$ (perfect fourth, Doc. 042/159) gives the leading digit, 3 (space dimensions) and $2^2{=}4$ (spacetime) are read off the geometry. The *magnitude-carrying* factor $5^4{=}625$ ($\rightarrow 10^{-4}$), however, is *not* derived forward: it is merely *asserted* ("emerges from the fractal structure", Doc. 009; "encodes the symmetry structure", Doc. 181 – there circular via $30000/4$) or the value is *empirical* (Doc. 042: measured from the Higgs sector $1{,}3194 \times 10^{-4}$, ~1 % off the ideal $4/30000$). It remains the *intuitive* (harmonic / Euler Tonnetz) resp. empirical grounding of the magnitude; this is openly *declared* as such, not presented as a forward derivation. Structural parallel to P20 (form fixed, factor open), one level deeper. *Declared/noted*

P34 — Doc. 181 / 013

Concerns: 181 / 013

Erroneous / unclear. $D = 4$ presented as "following necessarily / not a postulate"; role of the SI conversions for the c-/time-power

Correct / clarified. *3+1 is empirical input* – no theory derives the dimension count; the earlier "follows necessarily" claim (Doc. 181) is overstated. Forward are (i) periodicity (divergence-freedom), (ii) torus-not-sphere ($\pi_1 \neq 0$, stable windings), (iii) *minimal sufficiency* – 4 instead of a naive 5, because the duality $\tilde{T} \cdot m = 1$ lets one circle carry mass *and* time, (iv) the lossless (bijective) Hilbert representation (Type III, Doc. 270/206/230). The *bare number* 4 is *not* forced forward (the "4 from ξ" argument in Doc. 181 is circular resp. base-dependent). *Connection with the units:* the four-circle structure is at the same time the exponent bookkeeping of the SI conversions – c = space-time bridge (the three T^3 circles against the time direction), $\hbar$ = mass-time bridge (S^1_m, $\tilde{T}m = 1$). Hence the SI factors are indispensable for the *power of* c and the *power of time*: $c = \hbar = 1$ collapses L, T, M to one axis and destroys exactly the exponent structure one wants to read off. T^4 (mass reading) and T^3+time (time reading) are the same object. Signature as P20/P33: structure/form forward, number input. *Clarified*

P35 — Doc. corpus-wide esp. 182, 250s, sc

Concerns: corpus-wide (esp. 182, 250s, scale tables)

Erroneous / unclear. quantities of the form $X = \xi^N$ resp. $X = \varphi^N$ read as FFGFT confirmation; the SI conversion factor (ℓ_P) partly tacitly absorbed into the ξ exponent

Correct / clarified. ξ^N **triviality:** "$X = \xi^N$" is by itself *vacuous*, since any magnitude can be written as a ξ power – an exponent alone is not a prediction (cf. P10, P20, P33: form forward, number input). The SI conversion factor ℓ_P must *not* be absorbed into the exponent. **Finding:** only $L_0/\ell_P = \xi^1$ is exact; $R_H = \ell_P \cdot \xi^{-63/4}$ with $63/4 = 41/4 + 11/2$ – the corpus $41/4$ is referenced to E_0 instead of the Planck scale; the "nice fractions" are roundings of measured exponents (measured $\approx 15{,}72$ vs. $63/4 = 15{,}75$). A ξ^N representation counts as content only if N is derived forward. *Addendum (Doc. 279, 14 June):* the $11/2$ in $63/4 = 41/4 + 11/2$ is *not* *arbitrary* – it coincides with the forward-derived fine-structure exponent $\alpha = \xi E_0^2 = K\xi^{11/2}$ (Doc. 011/070, the same $E_0 = \sqrt{m_e m_\mu}$ as the $41/4$ anchor), to $\sim 0{,}5\,\%$ ($E_0/E_{\text{Planck}} = \xi^{5{,}48}$ vs. $5{,}5$). *A conjecture, not evidence*: different reference bases (MeV vs. Planck), the identity proven only by $E_0 = E_{\text{Planck}}\,\xi^{11/2}$ forward. The cosmic $41/4$ remains open. *Declared (changelog Update 16), table entry added 10 June 2026*

P36 — Doc. 182

Concerns: 182 / cosmol. sector

Erroneous / unclear. R_H interpreted as the "maximal scale" resp. size of the universe

Correct / clarified. R_H **interpretation refined:** $R_H = c/H_0$ is the *Hubble length* – the local expansion rate expressed as a distance – *not* the size of the universe and *not* the observable horizon (the latter $\approx 3\,R_H$ in ΛCDM). The actual universe exceeds R_H; only a *lower bound* is measurable. The "maximal scale" from Doc. 182 is tenable only as the upper scale of the *cosmological torus sector*, not as the extent of the universe. Consistent with P20 (R_H calibration as a declared external input). *Declared (changelog Update 17), table entry added 10 June 2026*

P37 — Doc. 009, 188, 271, 272, 275

Concerns: 009, 188, 271, 272, 275

Erroneous / unclear. φ in various documents in various roles: (a) phenomenological scaling factor between fractal levels (Doc. 188); (b) substrate geometry of HLV, not FFGFT (Doc. 271); (c) trivial representation $X = \varphi^N$ as a non-prediction (Doc. 272); (d) $\varphi^{12}/(1{,}314)$ with an empirical anchor (Doc. 009, P10)

Correct / clarified. Three levels clearly separated:
(1) Microscopic – input: $\xi = 4/30000$ (fractal dimension). φ is *not* active here; ξ and φ are different objects.
(2) Macroscopic – revised (see P38): $1/\varphi \approx 0{,}6180$ is a *pass-through point* of the ξ damping recursion $r(k+1) = r(k)\cdot(1-\xi)$ at $k^* = \log(\varphi)/\xi \approx 3609$ (the only fixed point: 0). The original derivation claim is downgraded to an open question – $k^*(c)$ exists for any target c, P35 applies (details: P38). The scale interpretation stands: levels with $q = 2/3$ internally convergent, no RG steps with factor 2.
(3) Particle physics – output: $Q_{\text{FFGFT}} \approx 0{,}6677$ (Koide scalar, Doc. 258) operates only on the lepton-mass scale. Not to be confused with $1/\varphi$.
Warning Doc. 272 / P35 / P10: the warning "any quantity is writable as φ^N – not a prediction" remains valid – and, after the revision of 11 June 2026, also *applies to* the k^* result from Doc. 275 (the earlier opposing delimitation here is retracted, see P38). *Declared; level (2) revised 11 June 2026*

P38 — Doc. 275

Concerns: 275

Erroneous / unclear. $k^* = \log(\varphi)/\xi \approx 3609$ presented as a "fixed point" result "without a free parameter"; delimitation from the φ^N triviality (P35) claimed with three arguments

Correct / clarified. Status clarification: (a) $1/\varphi$ is a *pass-through point*, not a fixed point – the only fixed point of $r \mapsto r(1-\xi)$ is 0. (b) $k^*(c) = [\log c - \log(D_f/3)]/\log(1-\xi)$ exists for *any* target $c \in (0,1)$ with identical parameter status: $k^*(Q_{\text{FFGFT}}) = 3029$, $k^*(1/2) = 5198$, $k^*(1/e) = 7499$,

$k^*(\alpha_{\text{em}}) = 36899$ – the free exponent was merely renamed k^* and solved for the target. P35 applies in full (first corpus-internal self-application of the guardrail). (c) The arithmetic ($r(3609) = 0{,}6179940$, error $4{\cdot}10^{-5}$) is correct; the observation would gain physical content only through an independent fixing of $k^* \approx 3609$ (level counting, observable) – currently open. *Revised after audit P. Stoychev, 11 June 2026*

P39 — Doc. 182, 267, 277

Concerns: 182, 267, 277, corpus-wide (H_0/R_H)

Erroneous / unclear. H_0/R_H calibration (P20/P36) readable as an FFGFT-specific gap resp. as an asymmetric disadvantage relative to ΛCDM

Correct / clarified. **H_0/R_H status refined (shared reference point, not an FFGFT gap):** the relation Planck length↔ R_H (exponent $41/4$, $R_H/L_0 = 6{,}49 \times 10^{64}$) is derived from first principles by *no* framework – why the universe is $\sim 10^{60}$ Planck lengths across is predicted by no theory.
ΛCDM fixes the relation via the *measured* H_0 – itself circular in the sense of the data degeneracy (Doc. 267). FFGFT adopts the established value for comparability and writes it in ξ notation. Hence the cosmic scale is a *shared empirical reference point of both models*, *not* a free parameter and *not* an FFGFT-specific gap.
Reinforcing: in FFGFT $\hbar, c$ are SI conversion and G is derived ($\xi \to \{m_e, \ldots\} \to G \to \ell_P$); a “measured” dimensional anchor in the standard-physics sense does not even exist here. The earlier picture “imported/to be cleaned up” (P20/P32) remains correct but is *not* to be read as a weakness relative to ΛCDM.
Remaining open point (symmetric): solely the *forward derivation* of the exponent $41/4$ from ξ (P20). It is missing – but it is missing for every framework, not only FFGFT. The only genuine model asymmetry is the *dark sector* ($\Omega_\Lambda, \Omega_{\text{dm}}$ as free densities), carried by ΛCDM alone (Doc. 277, cost table). *Declared 14 June 2026*

C6 — Doc. 004, 025, 030, 039, 041, 053, 061, 063, 068, 081

Concerns: 004, 025, 030, 039, 041, 053, 061, 063, 068, 081

Erroneous / unclear. Cosmological redshift presented as *wavelength-dependent* ($z(\lambda) \propto \lambda$, “UV stronger than radio”; e.g. Doc. 063 $z(\lambda_0) = \frac{\xi x}{E_\xi}\lambda_0$; Doc. 061 $\Delta z = 0{,}008 \ln(\lambda/\lambda_0)$; Doc. 068 $z(\lambda) = z_0(1 + \ln(\lambda/\lambda_0))$).

Correct / clarified. **Redshift is achromatic.** The fractal path-lengthening is purely geometric and stretches *all* wavelengths by the same factor: $1 + z = e^{\xi x}$, independent of λ. A *finite-element simulation* of the geometry yields no wavelength dependence; the leading term $z = \xi x$ is frequency-independent (Doc. 064). The observed cosmological redshift is achromatic – FFGFT gets this right, and the earlier chromatic “discriminator” drops out (it moves to the “fits” column, data degeneracy Doc. 267). Worked out in Doc. 280. *Declared 15 June 2026*

P40 — Doc. 005, 006, 011, 012, 133, 275

Concerns: 005, 006, 011, 012, 133, 275

Erroneous / unclear. Implicit: from ξ the *absolute values* (masses, α, G) follow directly and exactly; the *100* in K_{frak} and the *3609* of the φ-recursion are taken as related/fundamental.

Correct / clarified. Only ratios are exact. Absolute values carry the fractal/SI correction, absorbed into the bridge constants v (masses $m = r\xi^p v$), $E_0 = 7{,}398$ MeV (α) and $3{,}521 \times 10^{-2}/2{,}843 \times 10^{-5}$ (G; `calc_De.py`); in ratios they cancel ($m_\mu/m_e = 206{,}768$). Factored out $= K_{\text{frak}} = 1 - 100\xi$ (Doc. 011/012; Doc. 133: from D_f-consistency + RG over 17 orders, "not an adjusted correction"). *Keep separate:* **100** = RG amplification (Doc. 133); **3609** = depth at which $r(k) = \frac{D_f}{3}(1-\xi)^k$ passes the *passage point* $1/\varphi$ (Doc. 275) – not a fixed point, unanchored unless $1/\varphi$ is independently motivated (open, P37). Further couplings (backdoor) open; clean falsification only via forward prediction ($41/4 \to H_0$). *Declared 15 June 2026*

P41 — Doc. 267, 280, 221

Concerns: 267, 280, 221 (cosmological sector)

Erroneous / unclear. Redshift drift (Sandage–Loeb) proposed as a possibly sharp discriminator of static ($1 + z = e^{\xi x}$, $\dot{z} = 0$) versus expansion – a *new* observable rather than a re-reading of the same data, potentially piercing the degeneracy (Doc. 267).

Correct / clarified. Checked and not booked (degeneracy risk at the shared scale). ΛCDM signal $\Delta v = c\dot{z}\,\Delta t/(1 + z)$ over 25 years: $+6$ cm/s ($z{=}0.5$), sign change at $z_0 = 1.92$, -14 cm/s ($z{=}4$), -20 cm/s ($z{=}5$); ELT/ANDES order $\sigma \sim 2$ cm/s. The *amplitude* test is not degeneracy-proof: a secular evolution of the phase ξx at rate $s \approx (0.1\text{–}0.4)\,H_0$ mimics ΛCDM, and the natural FFGFT rate is $c/R_H = H_0$ *exactly* – both models share the scale (P36/P39); moreover the dipole systematic of the Galactic acceleration (≈ 17 cm/s over 25 yr, direction-dependent and subtractable) lies at the same order as the signal. Conditionally sharp remains only the *shape* test: the sign change at z_0, i.e. the two-band contrast $\Delta v(0.5) - \Delta v(4) \approx +20$ cm/s, cannot be mimicked by *any* z-constant secular channel – but it requires a corpus-level *advance* commitment to strict staticity ($\dot{z} = 0$) and low-z carriers beyond Lyman-α; time horizon ~2 decades. No falsification criterion booked; calculation: `python/Dok190_Skripte/drift_sandage_loeb_check.py`. *Declared 2 July 2026*

P42 — Doc. 282, 292

Concerns: 282, 292 (lepton sector)

Erroneous / unclear. Doc. 282 presented the pair $(\sqrt{2},\ 2/9)$ as *two pure numbers without a fit*; the precision analysis (Doc. 292) shows that the circulant with exactly this pair misses m_μ/m_e by $1.0 \cdot 10^{-5}$ at a measurement precision of $2.2 \cdot 10^{-8}$ – excluded as an exact pair at the pole-mass level ($\sim 450\,\sigma$ in the μ/e-precise direction).

Correct / clarified. One declared reference point: the ratio m_μ/m_e. The framework fundamentally keeps the purely theoretical derivation from ξ; the lepton sector receives – analogous to P39 (H_0/R_H as shared empirical reference) – exactly *one* declared reference point: m_μ/m_e fixes, at structural $r = \sqrt{2}$ (from $Q = 2/3$), the operative angle $\theta_{\text{ref}} = 0.2222220471$. The structural number $2/9$ is its *rational address* to seven digits ($|\theta_{\text{ref}} - 2/9| = 1.75 \cdot 10^{-7}$), not an exactness claim at the pole-mass level. The $450\,\sigma$ finding thus dissolves in the claim layer. Two delimitations belong to this: *first*, the status of ξ is untouched – FFGFT claims ξ as forced by the geometry; that dispute is not adjudicated here. *Second*, the lepton masses are in themselves *not an input of the theory*: the structural numbers ($\sqrt{2}$ from $Q = 2/3$, $2/9$ as address) stand on the theory side; the reference point belongs to the *comparison level* – it locates the geometric relation in the pole-mass/SI data (analogous to P39/P40: shared reference point resp. bridge constant) and would not be needed at all without the intent to compare. The honest price therefore lies not in the parameter count of the theory but in the *test balance*: m_μ/m_e is consumed as anchor and drops out as a test quantity. The gain: the m_τ prediction becomes input-error-free sharp, $m_\tau = 1776.9690 \pm 0.0001$ MeV ($+0.91\,\sigma$ against PDG 1776.86 ± 0.12; the Belle-II world average decides). For the bridge work (Doc. 291) the question sharpens accordingly: the dynamics must select θ_{ref}; $2/9$ remains its address and the pre-registered Stage-10 target (orbit resolution $\sim 0.1 \gg 10^{-7}$, unaffected). *Declared 2 July 2026*

P43 — Doc. 006, 011, 292

Concerns: 006, 011, 292 (`calc_De.py`)

Erroneous / unclear. Open question: whether the K_{frak}-with-100 formulas in the calculator `calc_De.py` need reworking after the generation-linear finding (Doc. 292, $N_g = g \cdot N_0$).

Correct / clarified. **No rework; recorded as a note.** Verified: `calc_De.py` does *not* use $K_{\text{frak}} = 1 - 100\xi$ explicitly. *(i)* $\alpha = \xi E_0^2$ with $E_0 = 7.398$ MeV (= the geometric route-2 value, Doc. 292); since $7.398/7.348 = K_{\text{frak}}^{-1/2}$, K_{frak} is already *absorbed* into the E_0 choice ($1/\alpha = 137.035$, no separate factor) – so the computation is already consistent with the two-route picture (Doc. 292), no contradiction. *(ii)* The masses run through per-particle (r, p) ($m = r\xi^p v$; e/μ/τ errors $1.18/0.66/0.37\%$) – *that* is the coarse ξ ladder in which the generation-linear law $N_g = g{\cdot}N_0$ (Doc. 292) applies. Replacing the (r, p) by N_g would be a gain only after $N_0 \approx 38.6$ is derived (otherwise fit against fit, P35); until then the (r, p) stand. No falsification/correction entry; the **100** in K_{frak} (fixed RG scale, P40) and N_0 (per-generation ladder factor) stay separate. *Declared 3 July 2026*

R41 — Doc. 284

Concerns: 284 (cf. 270, 249)

Erroneous / unclear. Doc. 284 (and the private Medin note) framed “every conversion to SI is itself a measurement” and “a geometry-specific memory cannot appear in any SI-carrying result” – too strong.

Correct / clarified. **The loss is not the SI conversion.** Per Doc. 270 the Type I duality decompactification $S^1_m \to \mathbb{R}_t$ ($\tilde{T} \cdot m = 1$) is a *mode-preserving embedding* (lossless); compactness survives as the discrete spectrum and as physical observables (T_{rev}, CHSH$\sim \xi$). The genuinely lossy steps are the *reverse* compactification $\mathbb{R} \to S^1$, the spatial projections $D3 \to D0$, and – for a read-out – the *finite, non-periodic window plus decoherence* (Test G). FFGFT *does* carry framework-specific SI observables: mass reading (masses/Koide/g–2), CHSH$\sim \xi$, D_f, L_0 (Doc. 270/249). *Narrowed claim:* the coherence-sector recurrence (back-flow $5{,}125$, Doc. 282) is not *resolvable* on a warm, finite-window SI-time observable such as EEG – not “absent from any SI result”. Doc. 284 (DE+EN) aligned accordingly. *Declared 19 June 2026*

R42 — Doc. 270

Concerns: 270 (cf. 248)

Erroneous / unclear. The selection of which T^4 circle becomes time was left implicitly open / treated as a bare empirical input.

Correct / clarified. **Not an open geometric question – it is the measurement.** In the compact view the four circles are symmetric (no space/time distinction); time is not selected by the geometry but is *relational* – the symmetry-breaking *is* the measurement (the Type I decompactification = read-out, Doc. 270/284). The embedded observer experiences one direction as duration (Doc. 248, self-referentiality). Bookkeeping closes with no gap: oneness of time = signature/predictivity (hyperbolic vs. ultrahyperbolic); *which* circle = the measurement / experienced situation; $3{+}1$ = the observer's situation, not a free parameter. New section added to Doc. 270 (DE+EN). *Declared 19 June 2026*

R43 — Doc. 285, 283

Concerns: 285, 283

Erroneous / unclear. Doc. 285 first stated the 3-vs-6 relation as "inequivalent structure / structural fork, not a coordinate artifact."

Correct / clarified. **Corrected to containment.** $C_3 < A_5$ (3-fold and 5-fold both permute the same icosahedron, `recompute_reconcile.py`) and $T^7 = T^4 \times T^3$ ($7 = 4 + 3$), so HLV ⊃ FFGFT at the group/dimension level – the models are reconcilable, not incompatible. The $r(\tau) \le 1$ result shows only that one cut-and-project map does not interconvert $\sqrt{2}$ and $2/\sqrt{5}$. The genuine difference is the *realised protected geometry*: FFGFT protects an independent 3-fold ($\sqrt{2}$, Koide $2/3$, confirmed); HLV the 5-fold ($2/\sqrt{5}$), and (Doc. 283) $\sqrt{2}$ is not among HLV's protected directions. Containment makes them compatible without identical. *Declared 19 June 2026*

R44 — Doc. 272, 190

Concerns: 272, 190 (P35), 285

Erroneous / unclear. Previously: φ had no distinguished role (P35: any quantity is writable as ξ^N *or* φ^N – a cautionary example with no predictive content).

Correct / clarified. **Sharpened (no contradiction).** φ acquires meaning *not* inside FFGFT but as the derived eigenvalue of HLV's 5-fold / perp sector (golden inflation $\varphi^2 = \varphi + 1$) – a marker of the boundary HLV\FFGFT. The same crystallographic restriction ($2\cos(2\pi/5) = 1/\varphi \notin \mathbb{Z}$) that forbids φ in FFGFT assigns it to HLV. P35 stands: φ as a fitting base carries no content, φ as a symmetry eigenvalue is structure, not a measured scale in an exponent – φ does not become an FFGFT prediction. Dot Theory classification (Doc. 272): φ = HLV-side discriminator of the structural nesting, entered as a contestable $O2$ rendering. *Declared 19 June 2026*

R45 — Doc. 285, 282

Concerns: 285, 282–284

Erroneous / unclear. Doc. 285 stated the d7 spectrum point too narrowly (read like "quasicrystal spectrum without information content").

Correct / clarified. **Sharpened.** A quasicrystal has two spectra: the *structure / diffraction* spectrum is pure point (sharp Bragg peaks, $\approx 21\times$ above a random null, `quasicrystal_information.py`) – that is the information and exactly HLV's substrate-distinguishability (stands); the *energy / Laplacian* spectrum is singular-continuous (no sharp levels) – hence no masses. P35 caveat is narrow: only fitting *one* target number as φ^N is content-free. Honest residual: null-reproducible was the *temporal* over-damped kernel (Doc. 282–284), not the spatial structure. *Declared 19 June 2026*

R46 — Doc. 296

Concerns: 296 (cf. 025, 026, 063, 156)

Erroneous / unclear. Doc. 296 §5 ("Speculation: unobservable intermediate ranges") books dark matter/energy as *possible signatures of unobservable intermediate-scale ranges* – formulated as open speculation.

Correct / clarified. Open check-point (not yet processed). This intermediate-scale formulation potentially runs *counter* to the already established corpus line, which treats dark matter/energy not as an open signature but as a *derived consequence*: Doc. 025 reads dark matter as a ξ-field effect and dark energy as "not required"; Doc. 026/063 hold that the universe does not expand and dark energy becomes superfluous (H_0 reinterpreted); Doc. 156 frames the entire observable reality as the projection of a single energy field with static redshift $z \approx \xi \ln(d/\ell_P)$. The *intermediate-scale* mechanism is a different explanatory route from the established ξ-field route and is not a load-bearing concern of the framework. To be reconciled in a later revision of Doc. 296: either align the intermediate-scale formulation with the ξ-field derivation or mark it explicitly as a subordinate additional consideration. It should also be noted that the observer/projection/emergence core of Doc. 296 is already established in the corpus (Doc. 156) and is not newly introduced there. No correction entry, no rebuild – booked as an open note. *Declared 6 July 2026*

R47 — Doc. 285, 297, 298

Concerns: 285, 297, 298 (cf. 286, 284, 295)

Erroneous / unclear. R43 booked "HLV ⊃ FFGFT dimensional" ($D7 = D4 + 3'$-window). Read in the *time sector*, this pure dimension count treated Marcel's spiral-time window as *additional* – as if it sat *beside* FFGFT's recursion time and enlarged HLV.

Correct / clarified. In the time sector: containment, not enlargement. In unified notation – both $D6$+time, Marcel as $D6$+spiral-time, FFGFT in Hilbert space as $D6$+recursion-time – the spiral-time window is not additive but *internal*: a bounded section (window) of the unrolled recursion time. Hence HLV ⊂ FFGFT *in the time sector*. Precisely, Marcel's object is the over-damped $-1/\varphi$ branch (the $3'$-window, Doc. 285/286: carries structure, no mass), *not* the full recursion operator R – R has two branches, the relevant one (φ, $D4$, masses, revival 5.125) is FFGFT's. The embedding ι (Doc. 297, open conjecture) is exactly the map realizing this window as a bounded projection of the compact recursion. *Distinction from R43 (no contradiction):* R43's ⊃ is the pure *dimension count* (the $3'$-window as additional); R47's ⊂ is the *derivation content in the time sector* (the window as a section of the named compact whole – FFGFT supplies the whole, HLV evaluates the section without naming it). Both readings coexist *as long as both are read in the same $D6$+time notation*. *OP8-consistent:* a warm, finite EEG window over a partial turn looks locally Archimedean (Doc. 295) and does not resolve the winding (R41, Doc. 284) – that U_3 adds nothing resolvable over U_2 is thus the *prediction* of the window embedding, not a failure. *Open (symmetric):* whether ι is *forced* (genuine HLV ⊂ FFGFT in the time sector) or merely *allowed* (compatibility) – forcing test like C3A5-BRIDGE2, on the time axis; ι remains a conjecture (Doc. 297). *Declared 8 July 2026*

R48 — Doc. 304

Concerns: 304 (cf. 295, 301, 303)

Erroneous / unclear. Doc. 304 stated the M_τ/M_B bridge to Marcel's Spiral-Time memory across scattered bookings; the *private covering note* to the first sending overstated ("real, not a benchmark artefact, unavoidable"). Marcel Krüger (author of the Spiral-Time Hypothesis) asked publicly that the boundary be made visible in the document itself and the bridge labelled a candidate rather than a secured mapping.

Correct / clarified. Note withdrawn, document made precise (not repaired). The overstatement lay solely in the *mail* and was withdrawn without reservation; the document was written throughout from the FFGFT reading and already booked the bridge as restricted. Doc. 304 gained a dedicated section "Provenance and the limit of the bridge": *(i)* two axes – the e-criterion itself is script-verified and derivable from ξ (*proven*, Doc. 295), only the *correspondence* to the operational M_τ/M_B is unsecured (logically *restricted*, Category B; empirically "no benchmark performed", i.e. without an independent, jointly sealed comparison); *(ii)* the operational M_τ/M_B definition (recursive access, semantic compression, affective weighting, identity binding, persistent state carry, predictive reuse) is untouched – curvature-fidelity is an *additional, FFGFT-side* diagnostic candidate, not the separator of that framework, and replaces none of those functions (both directions: $1/n$ texture without bio-functions possible; bio-memory not excluded by failing the e-criterion); *(iii)* symmetry – neither framework contains, grounds, explains, or subordinates the other, rejection leaves both unchanged; *(iv)* named provenance (M_τ/M_B, operational M_B, v0.3–v0.5 from Marcel Krüger's work; $D(k)$, e-criterion, FFGFT interpretation from the FFGFT corpus). The ledger-foreign term "untested" was avoided; the claim-state remains *restricted (Category B)*. No FFGFT result changed – only the framing of the relation to Marcel's objects. *Declared 11 July 2026*

R49 — Doc. 304, 305

Concerns: 304, 305 (cf. 295, 303, 272)

Erroneous / unclear. After R48 the provenance of 304 was deepened: Marcel Krüger asked publicly that the notation adopted from his Spiral-Time work (M_τ/M_B; likewise the sectors U1/U2/U3) be cited at first use — not only in the provenance section — and that the FFGFT-side objects optionally receive distinct notation.

Correct / clarified. Vocabulary cited, marked as analysis, abstract de-symbolised — notation not renamed. A consolidated first-use footnote attributes U1/U2/U3 *and* M_τ/M_B to Krüger (2026) with the primary source (`doi:10.5281/zenodo.21302278`) and documented lineage (Medinformatics; Smart Wearable Technology), and records: *FFGFT does not need this distinction* — a single, curvature-faithful winding; the sectors and the M_τ/M_B split serve only as an *analysis* of that framework, not as native structure. The Status separates accordingly: the FFGFT-internal core (thesis, linearisation reflex, measurement/projection epistemology) stands on its own, the correspondence is the illustrative, *restricted* overlay. Doc. 305 is booked in its Status as a *script-verified illustration* of Doc. 295 (not an independent finding; the uniform accumulator is a contrast foil from the bridge context, not an FFGFT object). The abstract was recast *narratively without the borrowed symbols*, so that M_τ/M_B and U1/U2/U3 first appear in the body (attributed). One sentence acknowledges the governance lineage (claim-state ledger / identity check from Stefaan Vossen's *Dot Theory*, co-developed in the IPI, cf. Doc. 272; the five-way record of Doc. 303 as the FFGFT-side extension). The *distinct renaming* of the FFGFT objects ($M_{\text{lin}}/M_{\text{curv}}$) was left as an authorial decision, not carried out. No FFGFT result changed — only citation, framing, and readability. *Declared 11 July 2026*

R50 — Doc. 025, 063, 061, 064, 078

Concerns: 025, 063, 061, 064, 078 (superseded by 306; cf. 003, 010, 295, 298, 302)

Erroneous / unclear. Docs. 025 and 063 justify the absence of a cosmic initial singularity by Heisenberg's energy–time uncertainty relation ("$\Delta E \cdot \Delta t \geq \hbar/2 = 1/2$ irrefutably proves…"). Three defects: *(M1)* **inverted implication** — a finite Δt gives $\Delta E \geq 1/(2\Delta t)$, a *finite* bound, not

$\Delta E \to \infty$; the latter follows only from $\Delta t \to 0$. *(M2)* **misapplication** — energy–time uncertainty is not canonical (Pauli: no self-adjoint time operator); its Δt is, by Mandelstam–Tamm, an observable's change time, not a cosmic age. *(M3)* **breach of style** — "irrefutably" is not a ledger term. Overarching: Heisenberg belongs to the *projected* QM level; an argument leaning on it does not leave the projection but circulates within it (Doc. 298).

Correct / clarified. Justification replaced, conclusion unchanged. The conclusion (no singular beginning; a finite core with a minimal scale L_0 set by ξ) stands. The derivation is replaced by the native chain of **Doc. 306**: $T \cdot m = 1$, hence with $E = m$, $T \cdot E = 1$ (Doc. 003/010) — an *equation*, not an inequality; from it $E \to \infty \Leftrightarrow T \to 0$; but FFGFT time carries a minimal, non-vanishing increment $d_1 = 100\,\xi_1 = 1/75$ (from $\xi = 4/30000$), so T is bounded below and E bounded above — an infinite density is *structurally* excluded. In addition the **boundary question**: $\partial T^4 = \emptyset$ (Doc. 302) — "what was at $t = 0$?" presupposes a boundary a compact torus does not have; the question is not wrongly answered but wrongly posed. Made precise: $D(k)$ *grows* without bound (logarithmically); the singularity-free statement is that $D(k)$ is finite at every *finite* k and diverges nowhere. **Reach:** the same chain (or its conclusion) also carries Doc. 061 ("the universe must have existed eternally"; $\Delta t = \infty$), Doc. 064 ("forbids a temporal beginning") and Doc. 078 ("forbids a classical Big-Bang singularity"). All named documents are *no longer authoritative* for this point — not refuted, but superseded. **The source documents are not revised:** the corpus is append-only, and this register entry is the authoritative correction layer, valid for all affected passages at once. Retroactively amending individual passages would in any case be incomplete (dependencies are semantic, not lexically findable) and would make overlooked passages appear falsely confirmed. Open (historical, not systematic, left unanswered): why the import was made although Doc. 003/010 was available. *Declared 12 July 2026*

R51 — Doc. 049

Concerns: 049 (cf. 095, 306, 267, 302; notation)

Erroneous / unclear. Doc. 049 ("Lagrangian comparison") contains two passages that *appear* to contradict the present position: "Big Bang as a field-excitation event" with $\delta m(x, t = 0) = \delta m_0\, \delta^3(x)\, e^{-H_0 t}$, and "Black holes as field singularities" with $\lim_{r \to r_s} \delta m(r) \to \infty$, $T(r) \to 0$. Read literally, the latter asserts a state of vanishing time — precisely the state Doc. 306 excludes ($T \cdot E = 1$ with minimal increment $d_1 = 100\,\xi_1 = 1/75$).

Correct / clarified. A notational relic, not a substantive contradiction — and nonetheless no longer authoritative. *(a) The resolution:* in the original T0 view, T_0 never denoted the instant zero but the *smallest possible time interval*. Doc. 095 uses it correctly as a reference scale: $\Omega(T) = T_0/T$. Read in that original sense, 049 means $T(r) \to T_0$, not $T \to 0$ — the limit towards the minimal time quantum. Hence $E = 1/T$ is bounded above by $1/T_0$: the *finite core with minimal scale L_0 (from ξ)* that 025/063 already name in their result, and exactly what Doc. 306 establishes natively. The contradiction is a misreading of the notation, not of the theory. *(b) The literal reading* ($T \to 0$, $t = 0$) is nevertheless *no longer authoritative*; Doc. 306 is. *(c) What remains superseded even on the charitable reading:* the "Big Bang" as an excitation event, the spatial point source $\delta^3(x)$, and the explicit H_0 in the exponent. The authoritative cosmological position is Doc. 306 (no singularity; no boundary: $\partial T^4 = \emptyset$, Doc. 302) together with Doc. 267 (no *metric* expansion, but $b = 1$; H_0 only *effective*, not metrically interpreted). *(d) Reach:* only the cosmological/astrophysical section of 049 is affected; the Lagrangian and particle sector of the same document (e.g. a_μ) is untouched. **The source documents are not revised** (append-only; this entry is the correction layer, cf. R50). *Declared 12 July 2026*

R52 — Doc. 101, 165

Concerns: 101, 165 (numerical values; cf. 180, 182, 013, 250, 290)

Erroneous / unclear. Two checkable errors in the H_0 context. *(i)* Doc. 101 (abstract and body) and Doc. 165 give the sub-Planck length as $L_0 = \xi \ell_P \approx 5.39 \times 10^{-39}$ m. *(ii)* Doc. 165 supports the ξ^{10} relation with the argument $L_H/L_\xi \sim \xi^{-10} \cdot$ (geom. factors).

Correct / clarified. Values corrected; the main relation is unaffected. *(i) Wrong L_0:* correct is $L_0 = \xi \ell_P = \frac{4}{30000} \cdot 1.616 \times 10^{-35}$ m = $\boxed{2.155 \times 10^{-39}\ \text{m}}$ — as in Doc. 180 (the dedicated derivation), Docs. 013, 181, 182, 187, 189, 268. The value 5.39×10^{-39} m implies $\xi = 3.34 \times 10^{-4}$ instead of $4/30000 = 1.333 \times 10^{-4}$ (factor 2.50); the mantissa 5.391 is that of the Planck *time* ($t_P = 5.391 \times 10^{-44}$ s), suggesting a transcription error. *(ii) Wrong pair of scales:* the supporting argument compares the Hubble length with the *granulation* scale L_0; computed, $L_H/L_0 = 6.37 \times 10^{64}$ while $\xi^{-10} = 5.63 \times 10^{38}$ — the required "geometric factor" would be 1.13×10^{26}, hence not one. The main relation in fact compares with the electron's *Compton* length, and there it holds: $L_H/\lambda_C(e) = 3.55 \times 10^{38} \approx [(\pi/2)\xi^{10}]^{-1} = 3.59 \times 10^{38}$. The supporting argument therefore fails; the result does not. *(iii) What stands:* the relation $\hbar H_0/(m_e c^2) = (\pi/2)\xi^{10}$ checks out numerically and yields $H_0 = 66.82$ km/s/Mpc (-0.9% against Planck's 67.4). *(iv) P35 unchanged:* the exponent 10 remains *not forward-derived*; the "structure argument" in 165 is a plausibility consideration, one leg of which (the scale ratio) does not compute. Untouched is the fact that a *separate* cosmological scale is structurally necessary (Doc. 182: no universal upper bound; each system carries $R_{\text{Torus}}(m) = \hbar/mc$) — P35 concerns the *notation*, not the substance. *(v) Doc. 306 unaffected:* its dense pole rests on the correct L_0; $E_{\text{max}} = \hbar c/L_0 = E_P/\xi = 9.16 \times 10^{22}$ GeV is confirmed (Docs. 250, 290). **The source documents are not revised** (append-only; cf. R50). *Declared 12 July 2026*

R53 — Doc. 267

Concerns: 267 §518 (table row *source of the scale*; cf. 182, 165, P20/P35/P36)

Erroneous / unclear. Doc. 267 §518 states: "FFGFT *derives* $E_H/\hbar \approx 66.2$ km/s/Mpc internally from $\xi^{41/4}$ — and hence R_H entirely from ξ (Doc. 182)." The symmetric-circularity table accordingly lists, on the FFGFT side, *source of the scale* as "$E_H/\hbar$ from ξ". Doc. 182, however, says the opposite — and is cited against its own booking: "The exponent $41/4$ is *not* derived from ξ but calibrated to the measured H_0 (P35/P36) ... not an independent ξ result." The same applies to the form $\hbar H_0/(m_e c^2) = (\pi/2)\xi^{10}$ (Doc. 165): the exponent 10 is likewise not forward-derived (cf. R52).

Correct / clarified. Corrected reading — and the symmetry becomes sharper, not weaker. *(a)* On the FFGFT side, the table row *source of the scale* correctly reads not "from ξ" but **"set / calibrated to H_0"**. *(b)* **Circularity is not a defect.** A measured angle θ_1 fixes only a length *ratio*; only the product $H_0 \cdot L_{\text{res}} \approx 458$ km/s is uniquely measurable. Every model must insert an absolute scale — the question is not *whether* but *where one enters the circle*. Doc. 267 already books this correctly: "Circular? Yes | Yes", and "the fair comparison is not ‚who is circularity-free' (nobody is), but parsimony against anchoring". *(c)* **The entry point must nevertheless be counted**, or the parsimony ledger is wrong: a calibrated exponent is an *input*, not a derivation. Writing it as $\xi^{41/4}$ or ξ^{10} makes a set quantity look derived and thereby hides that very input. *(d)* **The comparison remains valid and still favours parsimony:** FFGFT sets *one* declared sector scale; ΛCDM obtains r_s from six parameters plus a BBN early phase, and its H_0 is a pipeline output already presupposing expansion, FLRW and matter content (267 §518). *(e)* **P35 concerns the notation alone**, not the substance: that the cosmological sector requires its *own*, independently set scale is structurally necessary (Doc. 182: no universal upper bound; each system carries $R_{\text{Torus}}(m) = \hbar/mc$; cf. Doc. 306).

Recommended usage: declare the cosmological scale *openly as the sector's one external input*, rather than writing it as a power of ξ. $R_H \to \ell_1$ remains open (P20). **The source documents are not revised** (append-only; cf. R50). *Declared 12 July 2026*

R54 — Doc. 292, 306

Concerns: 292, 306 (cf. 307)

Erroneous / unclear. Two possible misreadings: (a) the ladder base unit $N_0 \approx 38.6$ (Dok. 292) might be mistaken for a *scale quantity alongside* the base units E_0, L_0, m_0, ξ_0; (b) the discreteness of time (Dok. 306) might be read as a *denial* of a common base tick.

Correct / clarified. **(a) N_0 is not a base unit.** $N_0 \approx 38.6$ is a *ladder-internal*, generation-counting factor ($N_g = g N_0$, Dok. 292); its value is *open* (P35: $100/e$, $100/\varphi^2$, 39 do not hit it reliably). It belongs neither among the base units nor among the reasons for the SI absolute-value deviation (these are: the SI conversion itself, the bridge constants $\nu, E_0, K_{\text{frak}}$, measurement precision; P40). **(b) There is a base tick.** Dok. 306 sets a minimal, ξ-fixed time increment $d_1 = 100\,\xi = 1/75$ — time *is* discrete. Since the modes always stand in exact relation through the fixed ξ-geometry, their individual cycles $T = \hbar/mc^2$ (from $\tilde{T} \cdot m = 1$) are commensurable multiples of the same increment: a *common base tick* (harmonics), not a continuum of free clocks and not an externally imposed world-time. Both precisions are consolidated in Dok. 307. *Declared 16 July 2026*

R55 — Doc. 292

Concerns: 292 §Cross-ladder coupling (cf. R54, P35, P40/P41)

Erroneous / unclear. The generation-linear correction law $N_g = g N_0$ is supported on three ladder entries ($N_1 : N_2 : N_3 = 1 : 1.981 : 2.971 \approx 1 : 2 : 3$) and quantified by a tolerance balance ("factor of about 300"). On recomputation the third entry carries no independent evidence, and the balance figure is not a second quantity. The identity $p_3 = p_1 + p_2$ is already named as information-free in the document ("carries no new information"); the consequence for N_3 and for the factor 300 is not drawn.

Correct / clarified. **The law stands; its evidential breadth is one entry, not three.** *(i) The third entry is forced arithmetically.* From $m_\tau/m_e = (m_\tau/m_\mu)(m_\mu/m_e)$ a uniformly multiplicative correction satisfies $K^{p_3} = K^{p_1} K^{p_2}$, hence $p_3 = p_1 + p_2$ *identically*; the stated accuracy $6 \cdot 10^{-14}$ is floating-point precision, not a measurement. Via $N_g = (1 - K^{p_g})/\xi$ this carries over to $N_3 \approx N_1 + N_2$: recomputed 115.95 against 115.55, a difference of $0.35\,\%$ and therefore second order in ξ. In the ratio $1 : 1.981 : 2.971$ *only the* 1.981 *is measured*; the third number is fixed as $1 + 1.981 = 2.981$. The raw deviations show the same additivity ($0.521 + 1.038 = 1.559$ against 1.565). *(ii) The factor* 300 *is the reciprocal of that same number.* A common N_0 imposes $N_1 = N_0$ and $N_2 = 2N_0$ simultaneously; this is satisfiable only to the extent that $N_2/N_1 = 1.9815$ departs from exactly 2 ($0.93\,\%$), and the residual factor is a priori of the reciprocal order ($\sim 10^2$). The tolerance balance therefore reports the residual of a one-parameter fit to essentially one observation. Note further that the raw deviations give $d_2/d_1 = 1.9923$ rather than 1.9815; this rounding difference shifts the expected residual factor by more than a factor of two — a further reason not to carry the 300 as a figure of merit. *(iii) The finding was declared and is then overtaken by the booking.* The preceding section ("K_{frak} counter-check") states: "The only striking point is that the τ/μ deviation is very nearly twice the μ/e deviation; whether structure or coincidence lies behind this remains unbooked under P35." That single observation carries the following section in full. The booking is to be brought into line with P35: the law is a *candidate resting on one observation*, not a threefold-supported regularity. *(iv)*

What is untouched. The mode separation (mode 1 reference-free against mode 2 under P42) is correct and is not affected. The finding that a *constant* K_{frak} cannot close the ladder while a single factor suffices for α (only one scale in play) remains valid and is independent of this entry; it is the load-bearing clarification of the role of the 100 (P40/P41). The value $N_0 \approx 38.6$ remains open as booked in R54. *(v) Note without claim.* Among the candidates for N_0 named in Dok. 292 ($100/e = 36.8$, $100/\varphi^2 = 38.2$, round 39), 39 is the only one with a provenance in the corpus: $\ln(10^{17}) = 39.14$ is the scale figure of the RG account of K_{frak} (Dok. 133/189/241). Neither passage cites the other. The departure from $N_0 = 38.65$ is $1.3\,\%$ and therefore does *not* meet the document's own standard ("do not hit it closely enough to be asserted"); recorded here as a cross-reference only, not booked as a candidate. *Verification script:* `python/Dok190_Skripte/r55_leiterkopplung_evidenzbreite.py`. **The source documents are not revised** (append-only; cf. R50). *Declared 19 July 2026*

R56 — Doc. 188

Concerns: 188 §Special role of $n = 3$ (cf. 189 §Derivation hierarchy, 248, 230/231/232, R55)

Erroneous / unclear. The three structural elements on which the construction rests are booked unequally in the corpus. $\tilde{T} \cdot m = 1$ stands in Doc. 189 as level 1 ("foundation") and in the text as "the only fundamental law"; T^4 and ξ are explicitly left open in Doc. 248 ("the first step — why T^4, why ξ — remains explicitly open"). Z_3, by contrast, appears in Doc. 188 as *selected*: it is said to be the "most parsimonious stable discrete solution". That argument alone does not select $n = 3$ — the same section states two sentences earlier that "stability increases with n"; parsimony and stability therefore point in opposite directions. What fixes $n = 3$ is the following sentence, via the number of observed generations.

Correct / clarified. The three elements are hereby booked uniformly as foundational assumptions. *(i) The declared framework.* FFGFT posits three structural elements: the fundamental relation $\tilde{T} \cdot m = 1$ (Doc. 189, level 1), the topology T^4 (Doc. 248), and the order Z_3 of the identification (this entry). They are stipulations, not derivations, and as such fully admissible; every theory enters its own framework somewhere. *(ii) Justifications remain motivation.* Where substantive reasons exist for a stipulation — for Z_3, the parsimony of the three-point configuration in the 2D cross-section and the agreement with the number of generations — they may be cited, but need not be: a stipulation does not become a derivation by the presence of reasons, and the absence of a written-out justification is no defect. Only the reversal of direction is to be avoided. The formulation in Doc. 188 that the three generations are "not free parameters but the three admissible Z_3 fixed points" reads the observation as a consequence of the symmetry; in fact the observation selects the symmetry. Recommended wording: Z_3 is stipulated, motivated by parsimony and consistent with the number of generations. *(iii) Unaffected:* once Z_3 stands, everything further is computation: $\theta = 2/9$ is the output of the Z_3-circulant diagonalisation on $L^2(T^4) \otimes \mathbb{C}^3$ and could have come out otherwise (Doc. 230/231/232; "found, not sought", Doc. 300). This entry concerns exclusively the layer above and devalues no result below it. *(iv) Consequence for external presentation.* "A single dimensionless parameter ξ" is accurate for the number of numerical parameters and remains so. The *structural* framework in which ξ sits is stipulated and declared. The complete statement — one numerical parameter, three declared structural elements, and here they are — is the more robust one, because it counts the entry point into the framework (cf. R53(c)). **Source documents are not revised** (append-only; cf. R50). *Declared 19 July 2026*

R57 — Doc. A015

Concerns: A015 (cf. 020, 030, A-Series)

Erroneous / unclear. The demand for a purely geometric derivation of $\xi = 4/30000$ was carried implicitly as an open point at several places in the corpus.

Correct / clarified. **Demand dissolved — ξ is a fundamental parameter.** A proof is neither possible nor necessary; the demand was wrongly posed. Five plausibility grounds are declared (with no claim of proof): (1) the fourth $4/3$ as a musical ratio in the Euler Tonnetz; (2) the fractal deviation $D_f = 3 - \xi$ close to the integer dimension; (3) scaling properties in the ξ-ladder formalism; (4) the prime-factor structure $4/30000 = 2^2/(2^4 \cdot 3 \cdot 5^4)$; (5) the music-theory connection (the author's workshop background). *Declared 22 July 2026*

R58 — Doc. A142, A140

Concerns: A142, A140 (cf. 127, 163)

Erroneous / unclear. Gravitational quantisation was carried at several places as an open point or implicit demand.

Correct / clarified. **Gravitational quantisation not required.** $G = \xi^2/(4m_{\text{char}})$ is intrinsic to the torus geometry; a separate quantum gravity does not arise, because the geometry T^4/Z_3 is pre-ordered and not dynamical in the same sense as quantum-field-theoretic fields. The quantum-gravity problem does not pose itself in this form. This is a point dissolved by declaration, not a solution of the problem. *Declared 22 July 2026*

R59 — Doc. A130

Concerns: A130 (cf. 011, 043, 101; P40)

Erroneous / unclear. The 2.3 % deviation between the ξ-derived α and the electroweak EFT precision value remained unexplained.

Correct / clarified. **Structurally explained [S].** Two causes: (i) SI values are scheme-dependent (1-loop, $\overline{\text{MS}}$), i.e. the EFT precision values are not theory-independent raw data; (ii) fractal corrections in v are not cleanly separable from the loop structure of the SM — v absorbs SM loops and fractal corrections jointly. Numerically: $\xi_{\text{EFT}}/K_{\text{frak}}^{3/2} \approx 1.330 \times 10^{-4}$; the remaining residue is $0.3\,\%$. This is a post-hoc structural explanation, not a proof; the layer marker remains [S]. *Declared 22 July 2026*

R60 — Doc. A160, A165

Concerns: A160, A165 (cf. 020, 022, 023, 230; P35)

Erroneous / unclear. The CHSH prefactor $\xi/(2\pi)$ and the cumulative formula $\text{CHSH}^{T0}(n) = 2\sqrt{2}\exp\big(-\xi \ln n/D_f\big)$ were not geometrically grounded in the corpus.

Correct / clarified. **Both factors now geometrically derived [B].** (i) 2π is the circumference of the unit circle — the natural conversion between the torus winding phase and the measurement angle; $\xi/(2\pi)$ is thus the phase correction per measurement. (ii) In the cumulative formula, $\xi \ln n/D_f$ integrates over D_f fractal dimensions; $\ln n$ counts the entanglement depth logarithmically, D_f in the denominator encodes fractal self-consistency. IBM Heron r2 (May 2026): $S = 2.74$ ($96.9\,\%$ of the Tsirelson value); device-to-device variation $\sim 1.4\,\%$ exceeds the ξ effect by a factor of ~ 1400 — not resolvable in principle with current NISQ hardware. *Declared 22 July 2026*

R61 — Doc. 174, 175

Concerns: 174, 175 (cf. A142, 147, 162; R57, P35)

Erroneous / unclear. Doc. 174 compares the measured T_2-ratio window (9–11 orders of magnitude, exciton vs. spin qubit) against *both* available ξ values and selects the one that fits: ξ yields 2 stages $= 7.75$ decades (rejected), $\xi_{\text{Higgs}} = 1.038 \times 10^{-5}$ yields 9.97 (accepted, "agreement $< 5\,\%$"). *After* this selection, the fitting value is declared "the right parameter for decoherence rates" and the two-spaces reading ("no contradiction") is supplied as justification; Doc. 175 additionally asserts the value "follows" from $\lambda_h^2 v^2/(16\pi^3 m_h^2)$.

Correct / clarified. Data-driven value selection — not a second parameter; withdrawn. *(i) The genesis:* ξ_{Higgs} was selected *because* it lands in the measured window; the two-spaces doctrine is rationalisation after the selection, not independent justification. Under P35 such an agreement carries no evidence. *(ii) The value does not match its own formula:* $\lambda_h^2 v^2/(16\pi^3 m_h^2) = 1.304 \times 10^{-4} \approx \xi$ (standard inputs $v = 246.22$ GeV, $m_h = 125.25$ GeV, $\lambda_h = m_h^2/2v^2$) — booked canonically as the Higgs matching of ξ itself (A142 [K]; deviation $2.2\,\%$, cf. R59). The printed value 1.038×10^{-5} instead corresponds to $\lambda_h^2 v^2/(64\pi^4 m_h^2)$ — an extra factor of 4π that is nowhere derived (match to $0.03\,\%$) *[Addendum 28 July 2026, cf. R66: since resolved — Doc. 310 identifies this factor as a miscount, $64\pi^4/16\pi^3 = 4\pi = S^2$, one 2-sphere too many; this confirms rather than weakens the present entry]*. The claimed Higgs-sector origin is therefore unsubstantiated as well. *(iii) Unaffected:* the stage-coupling law $C(N_1, N_2) \propto \xi^{|\Delta N|}$ (Doc. 175) with the *single* fundamental parameter (R57); the geometric-space / state-space distinction may stand as a figure of speech but carries no numerical parameter of its own. That no integer stage count hits the T_2 window with the single ξ (1 stage $= 3.875$ decades; 2 $= 7.75$, 3 $= 11.6$) is to be recorded as an open point, not closed by parameter choice. *(iv) Authoritative form for future use:* $\Gamma \propto \xi^{2\Delta N}\omega$ (amplitude $\propto \xi^{\Delta N}$, rate $\propto$ amplitude^2, golden rule) — declared and sealed in the SDCR Stage-2 pre-registration note v1.1 (SHA-256 `c47d73ce…3992`, 23 July 2026), *before* any stage-2 data exist. *Check script:* `python/Dok190_Skripte/r61_xi_higgs_matching.py`. **Source documents are not revised** (append-only; cf. R50). *Declared 23 July 2026*

R62 — Doc. 005

Concerns: 005 §Baryon worked example (proton) (cf. R50, P35)

Erroneous / unclear. The printed proton calculation $m_p = m_\mu \cdot K_{\text{corr}} \cdot QZ \cdot RG \cdot D_{\text{baryon}} \cdot f_{\text{NN}}$ with the printed values $0.105658 \cdot 0.985 \cdot 0.532 \cdot 1.00007 \cdot 0.363 \cdot 1.00002$ yields 0.0201 GeV, not the stated result 0.9381 GeV (missing factor 46.672). The formula line is moreover dimensionally open (m_μ in GeV, D_{baryon} in GeV $\Rightarrow$ GeV^2).

Correct / clarified. The anchor of the formula line is $\pi^2/2$, not m_μ. *(i) Diagnosis.* The missing factor identifies the error as a single symbol: $(\pi^2/2)/m_\mu = 46.705$ hits the missing factor to $0.07\,\%$. *(ii) Repair.* With $m_p = (\pi^2/2) \cdot K_{\text{corr}} \cdot QZ \cdot RG \cdot D_{\text{baryon}} \cdot f_{\text{NN}}$ — $\pi^2/2$ dimensionless, D_{baryon} (contains $0.5\,\Lambda_{\text{QCD}}$) the sole dimensionful carrier — the dimensional balance closes, and the *unchanged* printed values give 0.93878 GeV against PDG 0.93827 ($\Delta = +0.05\,\%$) resp. against the printed result 0.93810 ($\Delta = +0.07\,\%$). The printed values were internally consistent with the $\pi^2/2$ anchor; only the symbol m_μ in the formula line was erroneous. *(iii) Structural note.* $\pi^2/2 = \text{Vol}(B^4)$ is T^4-native; $(\pi^2/2)\cdot(16/\pi^2) = 8$ links the anchor to the reciprocal of the D4 packing density (A270). Further structure (the split $QZ \cdot K_{\text{corr}}$ underdetermined; geometric candidate $\pi/6$ for the product; $\pi^3/12$ link to the Higgs formula via the Weyl number 192) is kept independently in A155 §Baryon Candidate. *(iv) Secondary finding.* The printed value $K_{\text{corr}} = 0.985$ does not match the printed formula $K_{\text{frak}}^{D_f(1-(\xi/4)n_{\text{eff}})} = 0.9605$ ($\Delta = 2.5\,\%$); the worked example closes only with 0.985. Which value is correct is decided by the pending derivation. *(v) Untouched.* f_{NN} is an ML calibration factor (numerically ≈ 1). The meson formula of Doc. 005 is not affected by this entry; self-contained A-series successors are A155 (meson and baryon candidate) and A270 (sector structure). *Verification script:* `python/Dok190_Skripte/r62_baryon_anker.py`. **The source documents are not revised** (append-only; cf. R50). *Declared 23 July 2026*

R63 — Doc. 253

Concerns: 253 §ξ resonance heuristic (code: `rsa/factorization_benchmark_library.py`; cf. 024, 075, 076, 176; R57, P35)

Erroneous / unclear. Dok. 253 books the number-range dependence as "real but classical" and records that the ξ resonance heuristic favours $r \approx 2$ as the most frequent gcd case. The audited library further declares $\xi = 1/10$ "universally optimal" over $\xi_{\mathrm{FFGFT}} = 4/30000$. Both entries are correct but too weak: the cause of the optimality is not named, and the resonance function printed in Dok. 253 is not the one implemented.

Correct / clarified. ξ is inert in the "optimal" range; the optimisation switched the filter off. *(i) Inertness.* The code threshold is $1/1000$. For $\xi = 1/10$ and $\xi = 1/15$, *no* r in $2 \ldots 2000$ falls below it (0 of 1999); the first period found is always accepted. The algorithm is therefore output-identical to a ξ-free classical order search — identical output on 20 semiprimes, $19/20$ for both, and likewise for the absurd controls $\xi = 1/2$ and $\xi = 10^6$. *(ii) The fundamental value degrades performance.* With $\xi_{\mathrm{FFGFT}} = 4/30000$, 1998 of 1999 periods are rejected; $5/20$ successes remain, all of them from the `trivial_gcd` branch (base $2, 3, 5, 7$ divides n directly). From the period route: none. **The systematic "ξ optimisation" did not find an effective parameter value but the range in which the filter stops filtering.** Here an effective filter lowers the hit rate — which is why it was calibrated away. *(iii) The $r = 2$ maximum is constructive.* $\omega - \pi = \pi(2/r - 1)$ vanishes exactly at $r = 2$, for *every* ξ. The function selects no resonance but the smallest even order. The Dok. 253 entry is confirmed and sharpened: not "favoured" but maximal independently of the parameter. *(iv) Circularity of the strategy selection.* `_select_optimized_xi_strategy` calls, via `_categorize_number_optimized`, the trial division `_simple_factorize` ($d < 1000$). For all test numbers, n is thus already fully factorised *before* ξ is chosen ($20/20$); the categories `twin/cousin/distant_prime` presuppose p and q as known. *(v) Two different implementations of the same idea.* The Gaussian form $\exp(-(\omega - \pi)^2/(4|\xi|))$ stated in Dok. 253 describes the *HTML* tools correctly (`2/html/t0_shor_bigint.html`, `t0_Shore_simulator.html`); the *Python* library instead implements $1/(1 + |(\omega - \pi)^2/(4\xi)|)$ — Lorentzian, not Gaussian. Same maximum, incompatible tails. Both nevertheless end in the same finding by different routes: the Lorentzian form never bites at $\xi = 1/10$ (score at $r = 50$: $0.042 \gg 1/1000$), the Gaussian form underflows the floating-point representation for every ξ on offer. In addition, the Python version evaluates the resonance condition with $\pi \approx 355/113$ in exact `Fraction` arithmetic — $\omega = \pi$ is constructively unreachable except at $r = 2$. *(va) The HTML tools.* For *every* ξ that `t0_shor_bigint.html` offers (canonical $4/30000$; adaptive 10^{-4}, 10^{-5}, 10^{-6}, 10^{-7} by bit length), $\exp(-(\omega - \pi)^2/(4\xi))$ underflows to exactly 0.0 for *all* $r \neq 2$; only $r = 2$ yields 1. The `argmax` therefore falls on the first candidate, i.e. on the multiplicative order — classical order finding. Demonstrated: identical output over 20 semiprimes ($19/20$) for all five values on offer *and* for the controls 10^{-2} and 1. The adaptive bit ladder is thus without effect; `t0_Shore_simulator.html` shows the same degeneracy, its extra factor $(1 + \xi/r^2)^{2.5}$ likewise decreasing monotonically in r. *(vb) Test C in `t0_xi_num_algebraisch.html` can never fire.* The page computes $\xi_{\mathrm{num}} = \lambda(N)/N$ correctly (Dok. 253 books this rightly), but tests the ladder hypothesis with the criterion $|\log_{10} \xi_{\mathrm{num}} - \log_{10} \xi^N| < 0.3$. For *every* semiprime, $\xi_{\mathrm{num}} = \mathrm{lcm}(p-1, q-1)/(pq) < 1/2$, whereas the nearest window ($N = 0$, $\xi^0 = 1$) only begins at $10^{-0.3} = 0.5012$ and the next ($N = 1$) ends at 2.66×10^{-4}. The entire attainable range lies in the gap. The negative finding "ξ_{num} follows the FFGFT ladder: not proven" is therefore not a result about the data but a property of the test design — it could not have come out positive even had the hypothesis been true. *(vc)* `rsa/t0_factorization_demo.html` *does not compute.* The factors returned come from the lookup table `knownResults` (41 entries), success is rolled against hard-wired rates (`t0_optimized_*`: `success_rate 1.0`), and `calculateResonance` returns $(0.15+\texttt{random()}\cdot 0.25)\cdot$`success_boost` — a random number. `iterations`, `memory_mb` and `simulateExecutionTime` are random too. The page does not run a benchmark, it depicts one; it must not be cited as evidence. *(vi) What remains untouched.* All established facts of Dok. 253 remain valid, in particular "no global ξ optimises factorisation universally" — this entry supplies the mechanism. ξ as a fundamental parameter (R57) is unaffected: the finding invalidates no derivation, only an attempted application to the arithmetic number range whose bridging proof Dok. 253 already lists as missing. *(vii) Methodological consequence.* Any additional filter introduced in future — winding numbers, Z_3 cycles, higher resonance conditions — must carry, before application, an acceptance criterion that demonstrably *fails* for wrong parameter values. A filter whose selectivity is free to be tuned for hit rate optimises itself to zero. Pre-registration under P35 is not ornament here but the only safeguard. *Check script:* `python/Dok253_Skripte/r63_xi_faktorisierung_wirkungslos.py`. **The source documents are not revised** (append-only; cf. R50). *Declared 28 July 2026*

P44 — Doc. 253, 190

Concerns: 253, 190 (R63); cf. A135, P35

Erroneous / unclear. R63 shows that in every factorisation implementation audited, ξ sits inside the argument of a strictly monotone function of r and therefore cannot influence the selection *structurally* — a monotone reparametrisation moves no `argmax`. This left open the most charitable reading of the underlying idea: whether ξ acts as a *linewidth*, i.e. where there is scatter to filter and where the window is centred on the *measured* value rather than on $r = 2$. That question had nowhere been tested in the corpus.

Correct / clarified. Pre-registered, tested, and refuted for this design; the window width is a property of the measurement setup. *(i) Design.* Shor phase estimation, 12 counting qubits (phase resolution $2^{-12} = 2.44\times 10^{-4}$), 8 shots, $10\,\%$ fully random outcomes, 300 cases. *Reference A*: continued-fraction expansion of $k/2^n$ (standard procedure). *Variant B*: admissible are all r' for which at least the fraction `VOTE` of shots satisfies $\min_s |k/2^n - s/r'| \le w$; the smallest admissible r' is taken. Here w has a genuine role: too small and nothing is admissible, too large and $r' = 1$ or 2 slips through. An optimum exists. *(ii) Pre-registered criterion.* `VOTE` $= 0.6$ was fixed *before* the run; only w was swept. H1 counts as supported if (1) rate$(B, w = \xi) \ge$ rate$(A) + 2\,\%$, (2) $\arg\max_w$ lies within a factor 3 of ξ, and (3) the rate at $w = \xi/100$ and $w = 100\,\xi$ falls at least 10 percentage points below the maximum. *(iii) Result.* Reference A: $92.7\,\%$. Variant B at $w = \xi$: $72.7\,\%$; maximum $77.3\,\%$ at $w = 3.16\,\xi$; at $w = \xi/100$: $9.3\,\%$, at $w = 100\,\xi$: $23.3\,\%$. Condition (1) *not* met, condition (2) *not* met, condition (3) *met*. H1 is refuted for this design. *(iv) The actual finding lies in condition (3).* The filter *can* fail and does so at wrong values — which distinguishes this design from every one audited under R63, where the filter could not fail at any value. The difference is not the value of ξ but where the window sits. *(v) The width is set by the apparatus.* Across nine configurations (noise 0–$25\,\%$, 4–16 shots, $r_{\max} = 40$–250, 10–14 counting qubits) $\arg\max_w$ moves between $3.16\,\xi$ and $10\,\xi$ and tracks the phase resolution, not a constant. In *all nine* configurations $B_{\max} < A$ — the variant beats the standard procedure in none. *(vi) Placement relative to A135.* A135 books *settability to one* as a diagnostic of non-fundamentality: what can be normalised away was a property of the unit system. The case measured here is the same diagnostic in operational form, for an object to which the algebraic form does not apply: a fit parameter cannot be set to one, but it gives itself away when its optimum moves with the instrument. Both the precondition and the limit of the procedure is a *variable* setup — for the cosmological scale anchor (Dok. 309, inheritance problem) there is only one, which is why the same diagnostic does not apply there. *(vii) Estimate of researcher degrees of freedom.* If `VOTE` is varied after the fact, the location of the optimum moves: at `VOTE` $= 0.5$ it sits *exactly* on ξ, at 0.75 on $10\,\xi$. With two adjustable knobs instead of one the hit can be manufactured. This is precisely the mechanism documented under R63; hence `VOTE` is part of the pre-registration and immutable after the run. The script computes this degree of freedom itself without letting it enter the verdict. *(viii) What remains open.* The finding concerns the factorisation branch. It says nothing about ξ in the lepton sector, where the evidence rests on ratios and Koide and not on a tuned window (cf. R63 (vi)). ξ as a fundamental parameter (R57) is unaffected. *Check script:* `python/Dok190_Skripte/p44_xi_linienbreite_vorregistrierung.py`, SHA-256 `ce0126388269a105…7e91ca75`, seed 20260728. Any other seed is an independent test of the same criterion. **The source documents are not revised** (append-only; cf. R50). *Declared 28 July 2026*

R64 — Doc. 190

Concerns: `2/python/t0_no_fallback_shore.py`; B18 archive README; cf. R63, P44, P35

Erroneous / unclear. The file `t0_no_fallback_shore.py` is the only one in the corpus that keeps separate books for the T0 branch and the classical fallback, and its header states the reason explicitly: "no obscuring of the source of success". It is cited nowhere in the corpus; the sole mention sits in the B18 archive README and reads "deterministic Shor algorithm" — precisely what distinguishes the file is not named there. At the same time the code books successes of the T0 branch as `pure_t0_physics` resp. "T0-Energiefeld Physics", although R63 shows that this branch is classical order finding.

Correct / clarified. The procedure is right, the separating cut too coarse, and the label wrong. *(i) What holds.* The counters `t0_success_count` and `fallback_success_count` are incremented at separate places (l. 376 resp. 280/300), `get_statistics()` reports both plus `last_method`, and a built-in test function `test_pure_t0_vs_fallback()` exists. This is the method separation booked as missing in R63 (vii) — it was present and correct before it was needed. What was lost is not the code (it still sits under `2/python/`) but its use: the later libraries and the HTML tools no longer carry it. *(ii) The script logs its own refutation.* Across ten test cases the T0 branch prints 426 resonance values; *none* differs from 0.0. For $N = 14351$ the log reads "Best Period: r=144 (Resonance=0.000000)", followed by "Method: T0-Energiefeld Physics". The cause is the underflow of R63 (va); `max(periods, key=lambda x: x[1])` over an all-zero list returns the first element, i.e. the smallest period. *(iii) The cut runs along the wrong seam.* What is distinguished are the two *procedures* (T0 branch / fallback), not *parameter effective* versus *parameter ineffective*. Since the T0 branch is itself classical order finding, the existing bookkeeping separates two classical procedures from one another. *(iv) Where ξ actually acts in this file.* Twice, and both times as a *loop bound*: `max_check` = min($\sqrt{N}+1,\ 2\times10^{7}|\xi|/10^{-5}$) in the fallback (l. 287) and `search_range` = min($N,\ 10^{5}|\xi|/10^{-5}$) in the base selection (l. 311). At *fixed* search range the base selection returns the same base across six orders of magnitude of ξ — the weighting saturates. So ξ changes *how far* the search runs, not *what* is selected. The same pattern explains the 48-bit boundary documented in `technical_report.md` (R63). *(v) Recommended addition.* The existing separation is to be extended by a second level: a control run with altered ξ at *unchanged* loop bounds must yield a different result, otherwise ξ played no part in the result. That single line in the report would have made the R63 finding visible at once. For the success labels it follows that `pure_t0_physics` is to be replaced by a term naming the procedure ("order finding"), not its theory of origin. *(vi) What remains untouched.* The finding concerns naming and reporting depth, not the correctness of the factorisations — the factors returned are right. ξ as a fundamental parameter (R57) is unaffected. *Check script:* `python/Dok190_Skripte/r64_methodentrennung.py`. **The source documents are not revised** (append-only; cf. R50). *Declared 28 July 2026*

R65 — Doc. 024, 075, 076, 147, 176, 253

Concerns: 024, 075, 076, 147, 176, 253 as well as the entire `rsa/` folder and the HTML tools under `2/html/`; cf. R61 (pattern of withdrawal), R63, R64, P44, P35

Erroneous / unclear. In several places the corpus presents a "T0 factorisation procedure" resp. a "ξ period search" as an independent method distinguishable from Shor, and supports it with performance figures ($83.8\,\%$ and $97.1\,\%$ success rate, $O(1)$ resp. $O(\log n)$, "$\xi = 1/10$ universally optimal") as well as with a "number-range dependence of ξ".

Correct / clarified. No procedure distinguishable from Shor exists; the method claim is withdrawn. *(i) What is withdrawn.* (a) The claim of an independent T0 resp. ξ factorisation procedure. (b) The success rates $83.8\,\%$ and $97.1\,\%$ of the benchmark table. They are not measured; the $97.1\,\%$ moreover denote no factorisation but the harmonic classification of factors already found, which at ± 50 cents cannot fail (complete coverage of the octave by 63 target ratios). What was measured for the library is $O(\sqrt{n})$ runtime (growth $\times 3.8$ per $+4$ bits) — that is the trial division in `_find_factors`. *Not* objectionable is the column "Memory" with $O(1)$ / $O(\log n)$: it states memory use, not time complexity, and is substantially correct. The page makes no claim about time complexity. (c) `rsa/t0_factorization_demo.html` as evidence of any kind. (d) The "number-range dependence of ξ" as a finding; it is a hard-wired `if` cascade by bit length whose four branches behave identically. *(ii) Reason in one sentence.* In all five implementations audited, ξ sits either inside the argument of a strictly decreasing function of r — where it *structurally* cannot move the `argmax` — or as a loop bound, where it scales the search depth. Neither is a computational step. The four observed mechanisms by which the filter became ineffective differ and lead to the same result: threshold (Python library, R63 (i)), floating-point underflow (HTML tools, R63 (va)), window outside the attainable range (Test C, R63 (vb)), and complete coverage of the target space (± 50 cents, harmonic library). *(iii) What is retained and to be used further.* The code is a *correct classical Shor simulation*; the factors returned are right. Retained in particular: the BSGS order-finding in `t0_rational_optimized.py` (genuine $O(\sqrt{N})$); the restriction to even r (correctly halves the search, measured $\times 1.9$; it follows from the parity condition of the extraction, not from ξ); the exact `Fraction` arithmetic, whose advertised bit-identical reproducibility across hardware does hold; and the method separation of `t0_no_fallback_shore.py`, extended by the second level per R64 (v). *(iv) Binding terminology.* "T0 period search", "ξ resonance scoring" and `pure_t0_physics` are to be replaced by "order finding"; ξ in the loop bounds is to be named *search depth*. Any formulation claiming a performance of the parameter without an accompanying control run at altered ξ and *unchanged* loop bounds is to be avoided. *(v) What remains untouched.* **Shor's algorithm itself is not affected** — what was audited is solely what the implementations do in addition to it. Likewise untouched: ξ as a fundamental parameter (R57) and the lepton sector, whose evidence rests on ratios and Koide and not on a tuned window (R63 (vi)). This entry invalidates no derivation; it withdraws an attempted application to the arithmetic number range whose bridging proof Dok. 253 already lists as missing. *(vi) Authoritative form for future use.* What remains open is solely the line pre-registered in P44: ξ as the *linewidth* of a noisy phase estimate, with a window around the *measured* value. Negative for the design tested there (continued fractions remain superior, in all nine configurations), but it was the first test in this series that could have come out positive at all — the discrimination condition was met. Any resumption of the topic has to start there and not at the period selection. *Check* *scripts:* `python/Dok253_Skripte/r63_xi_faktorisierung_wirkungslos.py`, `r63a_html_werkzeuge.py`, `r63b_html_statisch.py`, `r63c_altarchiv.py`; `python/Dok190_Skripte/r64_methodentrennung.py`, `p44_xi_linienbreite_vorregistrierung.py`. **The source documents are not revised** (append-only; cf. R50, R61). *Declared 28 July 2026*

R66 — Doc. 097, 073

Concerns: 097 (DE/EN and the standalone versions under `2/tex-n/`), 073 (DE/EN); cf. 310, A142, A190, R59, R61, P35

Erroneous / unclear. The abstract of Doc. 097 states that it is "shown that the characteristic $16\pi^3$ structure in ξ is the natural result of rigorous quantum field theory", and announces a "complete mathematical derivation" carried out with "full mathematical rigour". Doc. 073 (DE/EN) cites this as "the QFT derivation via the Higgs sector (Doc. 097)". The *body* of Doc. 097, however, presents the same factor as two steps, the second of which is a choice of normalisation: $1/(16\pi^2)$ as the one-loop suppression and $1/\pi$ as "NDA normalisation", introduced with the sentence "many EFT authors use the rescaling $\xi_{\text{NDA}} = (1/\pi)\,\xi_{\overline{\text{MS}}}(\mu = m_h)$". Abstract and body of the same document therefore carry different claims.

Correct / clarified. **The $16\pi^3$ count is right — but not for the reason given in Doc. 097. Authoritative is the sphere count of Doc. 310.** *(i) The load-bearing reason is in Doc. 310* (§ "π^3 is geometry, not a loop" [K]): the factor is *not* a quantum loop factor but the product of the sphere surfaces of the compact T^4 boundary geometry,

$$16\pi^3 = (4\pi)\cdot(4\pi^2) = S^2 \cdot 2S^3,$$

verified numerically: $S^2 = 4\pi = 12.566371$, $2S^3 = 4\pi^2 = 39.478418$, product $496.100427 = 16\pi^3$ exactly. *(ii) The self-check of the count.* The earlier version $64\pi^4$ stands to the correct one as $64\pi^4/16\pi^3 = 4\pi = S^2$ — it counted *one 2-sphere too many*. That the error is exactly one whole sphere surface and not an arbitrary factor is the actual argument for the geometric count; a normalisation convention would not have to hit that value. *(iii) Earlier entries brought up to date.* C1 (Doc. 049) booked the correction $64\pi^4 \to 16\pi^3$ without a reason; R61 (ii) listed the factor 4π as "nowhere derived". Both held at the time of those bookings. Doc. 310 supplies the reason: the sphere count for C1, the miscount for R61. Both entries received a dated addendum on 28 July 2026; their wording is unchanged. The *withdrawal* in R61 is unaffected — it concerned the data-driven selection of ξ_{Higgs} as a second parameter, not the question of the π power; the miscount diagnosis confirms rather than weakens it. *(iv) The contradiction is between two documents, not in the number.* Doc. 097 grounds the factor in a loop plus an NDA rescaling; Doc. 310 states expressly that it is "not a quantum loop factor". Both routes end at the same number but are incompatible in content. Only one can bear weight, and as matters stand that is the geometric one. *(v) What is to be corrected.* The phrase "the natural result of rigorous quantum field theory" in the abstract of Doc. 097 is to be withdrawn; it claims as a result of calculation what its own body identifies as a choice of normalisation. Authoritative reading of Doc. 097: it shows that the expression is *also* formulable in EFT language — a plausibility reconstruction, not a proof of the π power. Doc. 073 is to be changed from "QFT derivation" to "Higgs matching". *(vi) Numerically unaffected.* Both booked forms are algebraically the same quantity: with $\lambda_h = m_h^2/(2v^2)$, $\lambda_h^2 v^2/(16\pi^3 m_h^2) = m_h^2/(64\pi^3 v^2) = 1.304005 \times 10^{-4}$ (A142, A190, Doc. 310), deviation from $\xi = 4/30000$: $2.2\,\%$ (cf. R59). *Check scripts:* `python/Dok310_Skripte/ffgft_310_pi_geometrie.py`, `ffgft_310_higgs_zugang.py`. **The source documents are not revised** (append-only; cf. R50). *Declared 28 July 2026*

C7 — Doc. 024, 075, 076, 176, 253

Concerns: the fourteen HTML pages under `rsa/` and `2/html/` that carried the finding booked under R63–R65; cf. R63, R64, R65, C1

Erroneous / unclear. Fourteen pages carried unmeasured performance figures, described classical order finding as a T0 procedure, imported a module that does not exist, or presented simulated results as measurements.

Correct / clarified. Rebuilt, not annotated: the pages now say what they do. No correction box was inserted; what was corrected is the computation, the labelling and the method text. *(i) Computation.* In `2/html/t0_shor_bigint.html` and `t0_Shore_simulator.html`, `t0FindPeriod` resp. `pureT0PeriodFinding` is now called `findOrder` and returns the multiplicative order — the smallest r with $a^r \equiv 1 \pmod N$. The resonance values are still computed but no longer enter the selection; instead the output shows how many of them are non-zero at all (measured: none). The method text gives the reason: for $r \geq 2$, $|\omega - \pi| = \pi(1 - 2/r)$ is strictly increasing, hence the resonance strictly decreasing — a monotone function moves no maximum, whatever the value of ξ. *(ii) Simulation replaced by computation.* `rsa/t0_factorization_demo.html` now computes: trial division, Fermat, Pollard rho, Pollard $p - 1$, order finding with Shor extraction, and for the harmonic library trial division followed by interval naming — all with real timing, real iteration counts and a deadline abort. Deleted are `determineSuccess`, `calculateResonance`, `simulateExecutionTime`, `generateFactors`, `simulateHarmonicTime` and `determineHarmonicSuccess`; `Math.random` no longer occurs in the computing part (previously: success, resonance, iterations, memory and runtime), nor do hard-wired `success_rate` values (previously eleven, among them `t0_optimized_*: 1.0`). `knownResults` remains as a category label for display; every result is checked against $p \cdot q = n$. *Consequence for the comparison table:* over 18 test numbers, order finding is about fifty times *slower* than trial division (12.5 vs. 0.21 ms) — which the simulated timings had concealed. *(iii) Labels.* "T0 Optimized / period finding" → "Order finding / smallest r with $a^r \equiv 1 \pmod N$"; "Harmonic hierarchical / ratio search" → "Harmonic library / trial division, then harmonic naming"; `pure_t0_physics` → `order_finding` at all five places including the scoring bonuses. In the tolerance table of `harmonic-factorization` the columns are now "Classifications" and "Classification rate" instead of "Successes" and "Rate"; the numbers $4/69$, $34/69$, $67/69$, $69/69$ remain — they are genuine measurements, only of the classification and not of the factorisation. *(iv) Figures and import.* $83.8\,\%$ replaced by "success depends on whether the order lies within the search bound"; $97.1\,\%$ in the benchmark table by $100\,\%$, because what works there is trial division, which always finds. `from t0_period_finding import RelativeT0` replaced by `from t0_rational_optimized import T0RationalSimulatorOptimized` — that is the class actually present with this function; module and class `RelativeT0` exist neither in the repository nor in the old archive. *(v) Test C.* `t0_xi_num_algebraisch.html` now first checks the reach of its own criterion and reports it. If the attainable range falls entirely into the gap between the windows, the verdict "test not informative" is issued instead of "hypothesis not supported". If it overlaps, the previous logic runs unchanged. *(vi) What was expressly not changed.* The column "Memory" with $O(1)$ / $O(\log n)$ in `libraries-benchmarks` is left untouched: it states memory use, not time complexity, and is substantially correct (cf. the correction to R65 (i)(b)). The measurements of the tolerance table and the `knownResults` categories likewise remain. *(vii) Verification.* All fourteen pages parse after the rebuild; the JavaScript syntax of the four computing pages was checked, and the computing functions were tested in isolation ($18/18$ correct factorisations per method, result identical for $\xi \in \{4/30000, 10^{-7}, 10^{-2}, 1\}$). *Rebuild script:* `python/Dok190_Skripte/k7_html_umbau.py` —

two-phase: all target sites are checked first, and nothing is written unless each occurs exactly as often as expected. *Declared 28 July 2026*

R67 — Docs. A040, A270, 133

Concerns: A040 (fractal correction), A270 (bulk exponent 36), Doc. 133 (older-corpus counterpart); cf. P35, R61

Erroneous / Unclear. The factor 100 in $K_{\text{frak}} = 1 - 100\xi$ is carried in A040 as a [K] "derived quantity", while its own text presents it as an approximation ("about 100", "geometric averaging"); no stated accuracy is given. The two booked notations also differ: $(D_f^{\text{eff}}/3)^{D_f^{\text{eff}}/2} = 0.986651$ against $1 - 100\xi = 0.986667$. Doc. 133 states the same in places more sharply ("amplification factor of 100" without "about"), likewise without an accuracy statement.

Correct / Clarified. *(i)* With $D_f^{\text{eff}} = 2.973$ (three decimals) the rounding interval leaves $n \in [98.3,\ 102.0]$ open; even $D = 2.973$ exactly gives $n = 100.12$. The stated accuracy is therefore about ± 2. *(ii)* The consistency condition fixing D_f^{eff} (m_e/m_μ fractal against $5\sqrt{3}/18 \cdot 10^{-2}$) itself carries $0.52\,\%$. *(iii) Consequence for A270:* the 0.098 difference between the two routes to the bulk exponent 36 is thereby covered – *no* real residual; $n = 100.273$ would bring coincidence. A270 additionally uses the naive $k^* = \ln\varphi/\xi = 3609.09$ instead of the exact $k^*_{\text{exact}} = 3608.51$ given in Doc. 275 (explaining 0.006 of the 0.098). *(iv) Propagation:* ± 2 on n gives $\Delta K_{\text{frak}} = \pm 2\xi \approx \pm 2.7 \times 10^{-4}$ ($0.027\,\%$); K_{frak} appears in the source of 18 of the 48 A-documents and enters every absolute value via A080 – ratios are unaffected by A040 [B]. *(v) Unaffected* are the assignment to the integer 36 (both routes stable, no value closer than 0.4 to a rounding boundary) and P35. *Check scripts:* `python/Dok190_Skripte/r67_faktor100_unsicherheit.py`, `r67_toleranz_k36.py` (addendum sections 2n and 7; the base check yields 17 hits better than $0.010\,\%$ against an expectation of ~ 16, rank of $74/75$ is $17/399$ – the hit is by itself not a signal). **Source documents unchanged** (append-only). *Declared 1 August 2026*

R68 — Docs. 279, 308

Concerns: 279 (H_0 chain), 308 (Λ reading, cosmic sector); cf. P20, P39, R53, R61; new: Doc. 312

Erroneous / Unclear. Doc. 308 correctly books Λ as a fundamental constant of nature as an artefact of the expansion reading, but leaves open what carries the Λ *function* instead. The status of FFGFT with respect to the Einstein equations was also undecided.

Correct / Clarified. *(i) Identification:* the Λ function is carried by the derived quantity $\Lambda^* := 1/R_H^2 = (\pi/2)^2\xi^{20}/\bar{\lambda}_e^2 = 5.218 \times 10^{-53}$ m^{-2} ("closure scale"). No new parameter. Doc. 308 stands: artefact as a constant, real as a derived scale. *(ii) Status label:* all links carry the new conditional label **[K|P20]**. *(iii) Field equations:* FFGFT adopts them *in full* (Lovelock forces the form including the Λ term); only the G value, the Λ value and the choice of solution remain open. Consequently the Eddington stability question applies and is not evaded; toy calculation (appendix 312): the winding/momentum equilibrium stabilises at $L^* = 2\pi R_H$ exactly if $T_w = \hbar c\Lambda^*/(2\pi)$ – the same step ξ^{20}, no second scale; momentum quantum at the minimum exactly $\hbar H_0$; tipping bound $N_{\text{Casimir}} < 12N_k$. *(iv) 10^{123} problem:* $\Lambda^* l_P^2 = 10^{-121.87}$ – the fine-tuning question reduces *entirely* to P20. *(v) Open, not fitted:* $\kappa = 2.12$; the $n{=}10$ candidates are marked post-hoc and are *not* bookable (R61 discipline). *Check scripts:* `python/Dok312_Skripte/` `ffgft_312_abschlussskala.py`, `ffgft_312C_stabilisierung.py`. **Source documents are not revised** (append-only; cf. R50). *Declared 3 August 2026*

R69 — Doc. 308

Concerns: 308 §a_0 derivation; cf. P20, R68; Docs. 306, 312

Erroneous / Unclear. Doc. 308 supports a_0 with *two separate* arguments (Unruh anchoring and the ξ^{10} chain); their structural connection remained unnamed.

Correct / Clarified. *(i)* From $\partial \mathrm{T}^4 = \emptyset$ (Doc. 306) an isotropic torus yields a compact time cycle $\tau_c = 2\pi/H_0 = 91.9$ Gyr [K|P20]; the energy quantum is exactly $\hbar H_0$, identical to the momentum quantum of the spatial cycles. *(ii)* KMS periodicity gives $T = \hbar H_0/(2\pi k_B) = 2.63 \times 10^{-30}$ K – exactly the Gibbons–Hawking temperature, here without a horizon [S]. *(iii)* Equating with Unruh gives $a = cH_0$ exactly: the compactness of the time direction is the *structural source* of the Unruh anchoring; two arguments become one (ratio 0.86 to RAR, unchanged). *(iv) New obligation D:* KMS holds for periodic *imaginary* time; identification with compact real time is a structural statement [S], obligation D(i). Closed timelike curves: no empirical conflict; the causality condition is periodicity of the field configurations, obligation D(ii) – coupled to the tipping bound $n_b - n_f/2 < 12N_k$. *Check script:* `ffgft_312D_kompakte_zeit.py`. **Source documents not revised.** *Declared 3 August 2026*

R70 — Docs. 061, A085 (T_0 scale gap)

Concerns: 061 (temperature/CMB), A085 (natural units), A080; cf. Docs. 279, 310, R61

Erroneous / Unclear. Three points. *(a) Scale gap:* A085 [B] requires every dimensionful constant to be "either unity or a calibration" – there may be only *one*. A080/A130 fix it: $E_0 = \sqrt{m_e m_\mu} = 7.348$ MeV, overdetermined by a second, α-free route. Doc. 061 instead uses the scale unit "1 eV": $k_B T_0 = (16/9)\xi$ eV hits to $0.92\,\%$ but requires $U = 0.9908$ eV; between U and E_0 there is no integer power of ξ ($\log_\xi = 1.77$). *(b) Notation clash:* in A085, T_0 in $m_0 c^2 = E_0 = k_B T_0 = \hbar/\tilde{T}$ denotes the *calibration temperature* (8.53×10^{10} K), in Docs. 061/313 the *CMB monopole* (2.7255 K) – a factor 3×10^{10} under the same symbol. *(c) Precision claim:* Doc. 061 states "$0.0002\,\%$" in its summary while its own detailed calculation reports "agreement to $1\,\%$"; the correct value is $0.92\,\%$.

Correct / Clarified. The CMB temperature is thereby the *only* quantity in the corpus without a Kammerton relation: $k_B T_{\mathrm{CMB}}/(m_e c^2) = 4.596 \times 10^{-10}$, $\log_\xi = 2.41$ (against E_0: 2.71) – no integer power of ξ. All other quantities (m_μ/m_e, E_0, α, $\hbar H_0$) trace back to the single anchor. A systematic search over rational prefactors yields only the chance density of hits (R61 control as with the base $74/75$ in R67); the task is therefore **forward derivation**, not search. Until then: Doc. 061 replaces the missing structure with a scale choice – a *calibration read as a derivation*. Either the relation must be converted to E_0 (changing its value), or the bridge between the two scales must be shown. *Unaffected* is the overdetermination of Doc. 313: it tests Ω_m, and the $0.92\,\%$ deviation propagates there as only 0.0003. **Source documents not revised.** *Declared 3 August 2026*

R71 — Doc. 309 (sharpening: side-taking in the Hubble tension)

Concerns: 309 (scale anchor, inheritance problem); cf. P20, P39, R69

Erroneous / Unclear. Doc. 309 names the calibration of exponent 10 to $H_0^{\Lambda\mathrm{CDM}}$ as the inheritance problem, but treats H_0 implicitly as a *number* one inherits.

Correct / Clarified. *(i)* There is no SI-anchored H_0 value: Planck 67.4 is the output of a six-parameter fit (no directly measured distance enters), SH0ES 73.0 comes from the multi-stage distance ladder; the discrepancy is $8.3\,\%$ (5–$6\,\sigma$, unresolved). FFGFT therefore inherits not merely a number but **the side-taking in the Hubble tension**, and that choice is not yet booked as a stipulation. Were SH0ES correct, the chain would require the prefactor 1.716 instead of $\pi/2 = 1.571$. *(ii)* The theoretical statement itself is SI-free: $\hbar H_0/(m_e c^2) = (\pi/2)\xi^{10}$ contains no unit; SI enters only at the comparison. *(iii) New open item:* the evidential weight of the chain rests on the *prefactor*. The ξ ladder is coarse (step spacing 3.875 decades); the chance that some step falls within $1\,\%$ is $0.22\,\%$ – significant *provided* $\pi/2$ stands independently. If the prefactor is free in the range 0.5–2, it covers $16\,\%$ of the step and the hit shows nothing. Hence to be settled and booked: **is $\pi/2$ derived or stipulated?** Until clarified, the H_0 chain carries its $0.86\,\%$ agreement with Planck as a *conditional* finding. *Declared 3 August 2026*

R72 — Docs. A010, A080, A265, A130 (E_0: bare and corrected)

Concerns: A010 (glossary), A080 (units), A265 (redshift), A130 (two E_0 routes); cf. A040, R67

Erroneous / Unclear. Three documents write the formula of one route with the numerical value of the other: "$E_0 = \sqrt{m_e m_\mu} \approx 7.398$ MeV", while the geometric mean gives 7.348 MeV. A130 carries both values as *two independent routes* without stating their relation.

Correct / Clarified. The two values are *not* two independent determinations but the same value **bare and fractally corrected**:

$$\frac{E_0(\alpha \text{ route})}{E_0(\text{masses})} = 1.006818, \qquad \frac{1}{\sqrt{K_{\text{frak}}}} = 1.006734 \qquad (0.008\,\%).$$

The square root is compelled because E_0 enters $\alpha = \xi E_0^2$ quadratically; the assignment follows A040 rule [B] (absolute values carry K_{frak}, ratios do not). Hence *(i)* the notation in A010/A080/A265 is abbreviated but structurally justified – the values remain unchanged and a note in each clarifies the assignment (bare $= 7.348$; α-anchored $= 7.398$); *(ii)* E_0 as calibration is unambiguous: the corrected value is carried, since only it matches α to 5×10^{-6} and an anchor must be *one* value; *(iii)* K_{frak} is **overdetermined**: the two E_0 values give $K = (E_1/E_2)^2 = 0.9865$, i.e. $n = 101.3$ – alongside $n = 100 \pm 2$ from the D_f^{eff} consistency (A040, R67) and $n = 100.27$ from the sector ladder (A270). Three independent routes, all within the R67 interval $[98.3;\ 102.0]$. *Incorporated 3 August 2026* (DE+EN).

R73 — Docs. 026, 279 (H_0 notations, exponent $41/4$)

Concerns: 026 (geometric cosmology), 279 (Casimir–CMB–H_0); cf. P20, P35, R68, R71

Erroneous / Unclear. The corpus carries three notations of the same H_0 chain without stating their relation; 279 additionally lists two points as open that are not: the exponent $41/4$ versus ξ^{10}, and the ratio $m_e c^2/(k_B T_{\text{CMB}})$ as an "anchor requiring a geometric explanation".

Correct / Clarified. *(i) The exponents are identical, not different.* The 279 form $L_\xi H_0/c = (\pi/2)(15/\pi^2)^{1/4}\,(m_e c^2/k_B T_{\text{CMB}})\,\xi^{41/4}$ and the Kammerton form $H_0 = (\pi/2)c\xi^{10}/\bar{\lambda}_e$ (Doc. 312) both give 66.821 km/s/Mpc (ratio 1.000000). $L_\xi = (15\xi/\pi^2)^{1/4}\hbar c/(k_B T_0)$ itself carries $\xi^{1/4}$; on substitution $\xi^{41/4}/\xi^{1/4} = \xi^{10}$ cancels. $41/4$ is the notation with the L_ξ anchor, 10 the one with the Kammerton – no second exponent and no second calibration. The P20 question is unaffected. *(ii) The ratio $m_e c^2/(k_B T_{CMB}) = 2.176 \times 10^9$ is not an anchor.* Under the same substitution it cancels *completely*, as does $(15/\pi^2)^{1/4}$; what remains is $(\pi/2)c\xi^{10}/\bar{\lambda}_e$. It is a pass-through term of the L_ξ notation and therefore requires no geometric explanation. The open point booked in R70 concerns something else – what sets T_0 itself – and stands. *(iii) A real deviation of $0.96\,\%$:* Doc. 026 carries a third chain, $H_0 = E_0\,\xi^{41/4}/\hbar$ with the E_0 anchor, obtaining 66.18 instead of 66.82 km/s/Mpc. The factor is $(E_0/m_e c^2)\,\xi^{1/4}/(\pi/2) = 0.9904$. *(iv) Resolution* *of the $\pi/2$ question of R71:* $\pi/2$ is *not a prefactor*. Written as a length instead of a rate, π disappears entirely: $L^* = 2\pi c/H_0 = 4\,\bar{\lambda}_e/\xi^{10}$ (exact to ten digits; Doc. 312). The primary statement is: *the cycle is four Kammertöne times the ladder step*; the $\pi/2$ arises solely from $H_0 = 2\pi c/L^*$. The objection of a freely choosable order-one prefactor thus lapses – in its place stands an integer. *(v) What remains open:* the origin of the four (torus directions, 2^2, D4 – all post-hoc, not bookable under R61) and the $0.96\,\%$ of the 026 chain, which in this language requires 3.9616 instead of 4.000 Kammertöne. An integer against a non-integer: the deviation is to be sought in Doc. 026, not in the Kammerton chain. *Check script:* calculation in `python/Dok313_Skripte/ffgft_313_fraktale_wegkorrektur.py` (section 7 and appendix). **Source documents unchanged.** *Declared 3 August 2026*

R74 — Docs. 025, 063 (lithium problem: mechanism, not solution)

Affects: 025 (Cosmology), 063 (Cosmic), DE+EN each, table "Cosmological problems: standard vs. T0"; cf. Doc. 267, Doc. 313 (Section F.4)

Overbooked. Both documents list the lithium problem in a table whose third column is headed **"T0 solution"**; the entry reads "nucleosynthesis over unlimited time". The claim level is one step too high: the mechanism named addresses the *window duration* and is consistent with the static reading, but from "no fixed time limit" **no numerical value** for ^{7}Li/H follows. It is a sketch of a mechanism, not a quantitative derivation.

Clarification. By the robustness classification in Doc. 313 (Section F.4), ^{7}Li/H belongs to the class of *number and duration quantities*: it depends on $\eta = n_B/n_\gamma$, on the window duration, and – in the measurement itself – on stellar atmosphere models; three model layers stacked. Such quantities are not robust under the criterion (no scale bridge, no duration, same level). The standard picture's factor-3 discrepancy therefore shows that something in the chain is wrong – *not where*. It is consequently no clean test *against* ΛCDM either.

Reach of the correction. Affected is only the claim level of the lithium row, neither the mechanism nor the other rows of the table. A future version should carry the wording as "mechanism approach" rather than "solution", with the explicit note that a quantitative ^{7}Li prediction is outstanding. The same check found that **no other older document** presents BBN observables (Y_p, D/H) as a passed test; Doc. 267 already books the BBN gap correctly as an open research question.

Symmetry of the finding. The classification does not exempt the other side: in ΛCDM too, Y_p is no model-free test and η comes from a multi-parameter fit. Doc. 267 already records that neither framework is circle-free; the fair comparison is parsimony against anchoring. *Check script:* `python/Dok313_Skripte/ffgft_bbn_skaleninvarianz.py` (Part C). **Source documents unchanged.** *Declared 5 August 2026*

R75 — Docs. 024, 034, 075, 076, 147, 176: ξ name collision and rewrite of the Shor documents

Finding. Six documents used the symbol ξ for two independent quantities: the FFGFT geometry parameter $\xi = 4/30000$ and a numerical search parameter of the factorisation algorithms (values $1/42$, $1/100$, $1/1000$, $1/100000$). The two are unrelated. Further misattributions followed from this collision.

Correction. In all affected documents the search parameter is now denoted σ and placed as a sampling grid without physical meaning. Doc. 176 additionally introduces three strictly separated parameter spaces: geometric ($\xi = 4/30000$, fixed), algebraic ($\xi_{\text{Higgs}} \approx 1.038 \times 10^{-5}$) and number-theoretic $\mathbb{Z}/N\mathbb{Z}$ (*no* universal parameter; the period r is determined by lcm$(p - 1, q - 1)$).

Substantive corrections.

- *Resonance principle retained, carriers corrected:* period finding is a spectral problem, but its generators are the **primes**. The bicoherence test (Doc. 316) shows coupling exactly at prime powers ($b^2 = 0.76$–0.85), none at composite numbers ($b^2 = 0.018$–0.025). All σ values have composite denominators and form mixed points without a coupling line.
- *Scaling measured:* the mean multiplicative order grows with exponent 0.95 – effectively $O(N)$. The $O((\log N)^3)$ claim for the classical method is thereby refuted empirically; it holds only for Shor's quantum method.
- *Winding number instead of frequency ratio:* closure is decided by $W = \log_2(r)$, not r. Closure is an artefact of rationalisation.
- *Two limits of different nature:* the Weyl obstruction ($N(\lambda) \sim \lambda^{d/2}$ against $N(T) \sim T \ln T$) concerns the representation of *infinite* spectra. Shor's task is finite; it is limited by gate count (n^3) and error-correction overhead, not by Weyl. Invoking the Weyl limit against Shor would overextend it.
- *Thermal advantage of optics:* at 600 nm and 300 K the thermal occupation is 1.8×10^{-35}, at 5 GHz it is 1250. Superconducting systems therefore require $T < 52$ mK; optical systems require none. The advantage concerns infrastructure, not computational power.
- *Exponential dimension:* follows from the tensor product structure, not from the carrier medium. A single photon in n modes gives the same dimension n as a classical wave.
- *Entanglement:* does not carry the computational advantage on its own – Gottesman-Knill shows maximally entangled Clifford circuits to be classically simulable in polynomial time. Which quantity carries it is open and is named as open.

Verification. Three scripts in `2/python/Shor_Skripte/`: `s1_periodensuche.py`, `s2_grenzen.py`, `s3_thermik_zustandsraum.py`.

Scope. Rewritten: Doc. 075 (German version), Doc. 076 (DE+EN), Doc. 176 (DE+EN). Revised: Doc. 147 (DE+EN; the IBM Heron r2 measurements of May 2026 remain unchanged and load-bearing), Doc. 024, Doc. 034 (DE+EN). Note: `075_RSA_En` is a different document (general RSA cryptography without T0 reference) and is not affected. As an interim solution the English translation of the rewritten German version carries the filename `075_T0_Period_En_ch.tex`; the wrapper is `075_T0_Period_En.tex`.

Addenda. The new cost table in Doc. 176 establishes that preparation accounts for approximately $95\,\%$ of gate cost and the QFT for only $5\,\%$; this figure has been added to Docs. 075, 076 (DE+EN), and 147 (DE+EN). The description of Doc. 176 in Doc. 173 has been updated to reflect the new content.
Declared 18 August 2026

R76 – Scope of mass derivations (August 2026): Doc. 318 explicitly delimits the scope of ξ mass derivations. Derived: lepton rest masses [K]. Open bridge: hadron masses (QCD confinement contribution not derived from ξ). m_p/m_e is not a derivation target of the current corpus. Three scenarios for the QCD bridge are marked as research questions.
Declared 20 August 2026

R77 – Classification of early coupling constant formulas (August 2026): The formulas in Doc. 041 for the strong, weak, and gravitational couplings ($\alpha_s = \xi^{-1/3}$, $\alpha_W = \xi^{1/2}$, $\alpha_G = \xi^2$, $\Lambda_{\text{QCD}} = v \cdot \xi^{1/3}$) are *structural sketches in natural units*, not SI precision derivations. They show the principal ξ-dependence, not the exact SI value. In particular, $\xi^{-1/3} = 9.65$ in natural units; the unit bridge to $\alpha_s(M_Z) = 0.118$ ($M_Z = 91.19$ GeV, mass of the Z boson, reference scale for the strong coupling) is not carried out in Doc. 041. Status of all formulas in Doc. 041: [S] (structural sketch). The precision derivation of the strong coupling is in

Doc. 160: $\alpha_s(m_\tau) = 3\xi^{1/4} \approx 0.322$ (deviation $-2.3\,\%$, [K]). Doc. 318 (R76) contains the complete classification.
Declared 20 August 2026

R78 – α_s and running coupling (August 2026): The strong coupling constant α_s is a running coupling: $\alpha_s(m_\tau) \approx 0.33$ and $\alpha_s(M_Z) \approx 0.118$ are different values of the *same* constant at different energy scales. Doc. 160 derives $\alpha_s(m_\tau) = 3\xi^{1/4}$ geometrically [K]. Doc. 005 uses $\alpha_s(M_Z) = 0.118$ as SM input for the proton mass calculation – consistent because M_Z is the relevant scale there. Open: the RGE bridge (renormalisation group equation) from $\alpha_s(m_\tau)$ to $\alpha_s(M_Z)$ within FFGFT is not yet carried out [S].
Declared 20 August 2026

R79 – Algebraic derivation of the $SU(3)_c$ gauge group (August 2026): Doc. 321 closes the open bridge marked [S] in Docs. 319, 320 and 322. The eight Gell-Mann operators are constructed algebraically from the three $\mathbb{Z}_3$ eigensectors $\mathscr{H}_0, \mathscr{H}_1, \mathscr{H}_2$ of $L^2(T^4)$; their $\mathfrak{su}(3)$ commutation relations are proved with canonical structure constants. $N_c = 3$ is algebraically necessary, not postulated. Confinement is triality selection $T_R = 0$ (algebraic form of Doc. 049). Status: **[B]**.
Declared 22 August 2026

R80 – $SU(2)_L$, $U(1)_Y$ and $\sin^2\theta_W|_{\text{GUT}} = 3/8$ (August 2026): Doc. 321 additionally derives: $U(1)_Y$ from the fourth torus direction T^1 [B]; $SU(2)_L$ from the $\mathbb{Z}_2$ pairing $\mathscr{H}_1 \leftrightarrow \mathscr{H}_2$ [B]; $\sin^2\theta_W|_{\text{GUT}} = 3/8$ from the trace formula of the five orbifold states $(d_R, d_G, d_B, e^-, \nu_e)$ [B]. The classical $SU(5)$ GUT result (Georgi-Glashow 1974) is derived from FFGFT geometry. Status: **[B]**.
Declared 22 August 2026

R81 – Weinberg angle at M_Z from ξ (August 2026): Doc. 323 closes the RGE bridge declared [S] in R78 and Doc. 321. The complete 1-loop derivation gives

$$\sin^2\theta_W(M_Z) = \frac{3}{8} - \frac{55\,\alpha_{\text{em}}(M_Z)}{24\pi}\left[\ln\frac{m_{\text{Pl}}}{M_Z} + \frac{19}{12}\ln\xi\right] = 0.2308 \quad (\text{PDG: } 0.2312, -0.19\,\%).$$

Inputs: $\alpha_s(m_\tau) = 3\xi^{1/4}$ [K] (Doc. 160), $N_c = 3$ [B] (Doc. 321), m_τ and M_Z from ξ [K] (Doc. 320). The exponent $p = 19/12 = p_e + 1/(4N_c)$ is geometrically grounded ($p_e = 3/2$: electron mass exponent, Doc. 320). The RGE bridge declared [S] in R78 is now **[K]**. Remaining external inputs: m_{Pl} and $\alpha_{\text{em}}(M_Z)$ (open [S]).
Declared 22 August 2026

R82 – Hilbert-space embedding and spectral theory (August 2026): Doc. 322 develops the mathematical foundation of FFGFT spectral theory. The fundamental fractal operator $\hat{F}$ is defined on $\mathscr{H}_{\text{FFGFT}} = \mathscr{H}_{\text{geom}} \otimes \mathscr{H}_{\text{spin}} \otimes \mathscr{H}_{\text{flavor}}$. $\xi = \lambda_{\text{min}}(\hat{F}_{D_4})$ is the smallest eigenvalue of the D_4 sub-operator [K]. Fixed-point boundary conditions $\psi_\chi(x_0 + y) = \chi \cdot \psi_\chi(x_0 + g_*(y))$ ground the neutrino localisation [K] (cf. Doc. 320). The fractal measure with 100-fold recursion secures $D_f = 3 - \xi$ [K]. Open: complete self-adjointness proof for $\hat{F}$ [S].
Declared 22 August 2026

R83 – Status update: closed bridges (August 2026): The following open bridges previously declared [S] are closed by Docs. 319–323:

- $SU(3)_c$ emergence from $\mathbb{Z}_3$ triality: **[B]** (Doc. 321, R79).
- $N_c = 3$ algebraically necessary: **[B]** (Doc. 321, R79).
- $SU(2)_L$ and $U(1)_Y$ from orbifold geometry: **[B]** (Doc. 321, R80).
- $\sin^2\theta_W|_{\text{GUT}} = 3/8$: **[B]** (Doc. 321, R80).
- Weinberg angle $\sin^2\theta_W(M_Z)$ from ξ: **[K]** $-0.19\,\%$ (Doc. 323, R81).

- RGE bridge to $\sin^2\theta_W(M_Z)$ (R78): **[K]** (Doc. 323, R81).
- Fixed-point boundary conditions on $T^4/\mathbb{Z}_3$: **[K]** (Doc. 322, R82).
- Gell-Mann matrices from orbifold modes: **[B]** (Doc. 321, R79).

Remaining genuine open bridges: Δm^2_{32} ($-22\,\%$, Doc. 320 [S]); self-adjointness of $\hat{F}$ (Doc. 322 [S]); 2-loop corrections to α_s [S]; m_{Pl} and α_{em} from ξ [S]; quark sector (Doc. 318).
Declared 22 August 2026